

BFS Traversal

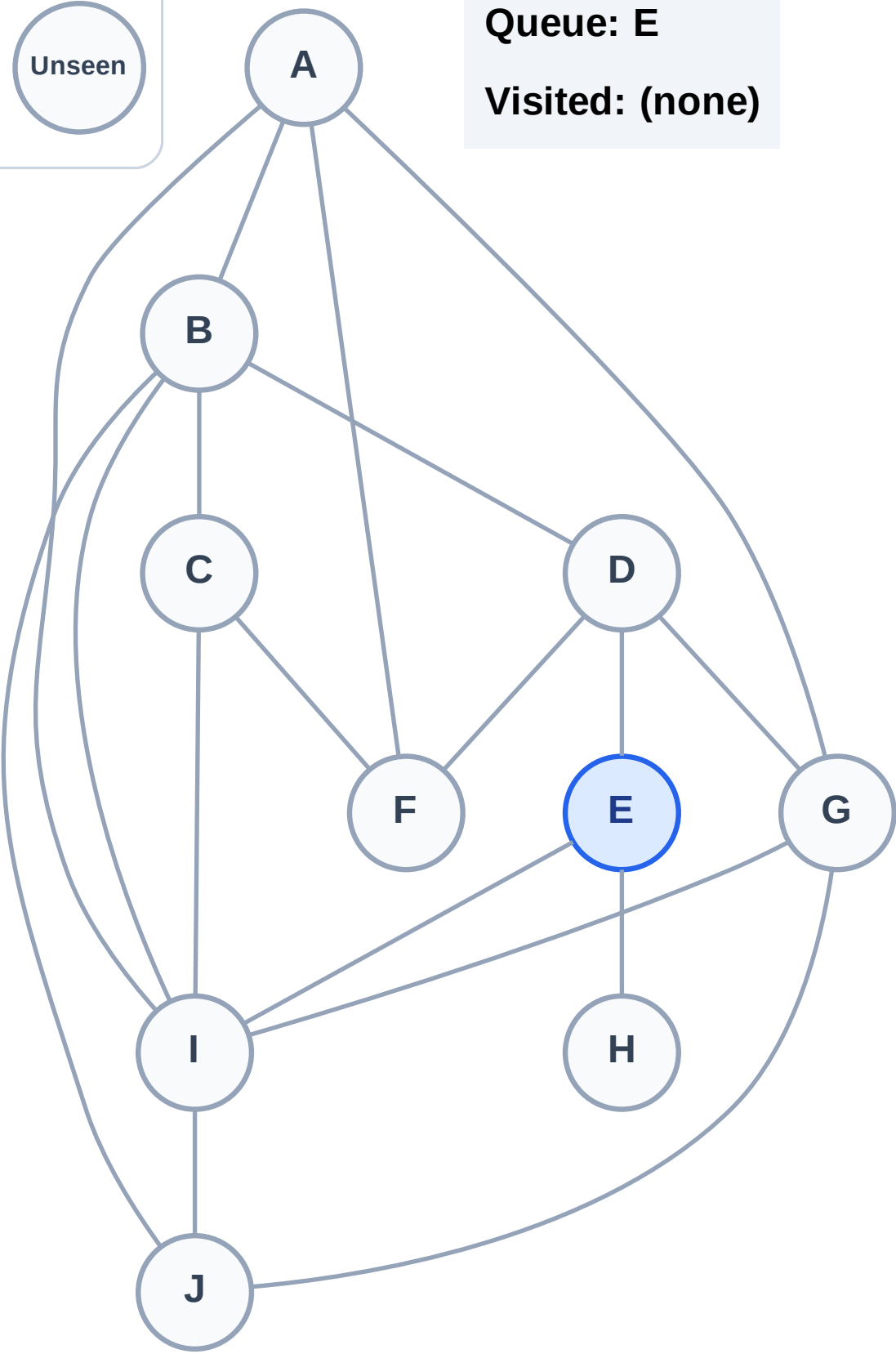
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

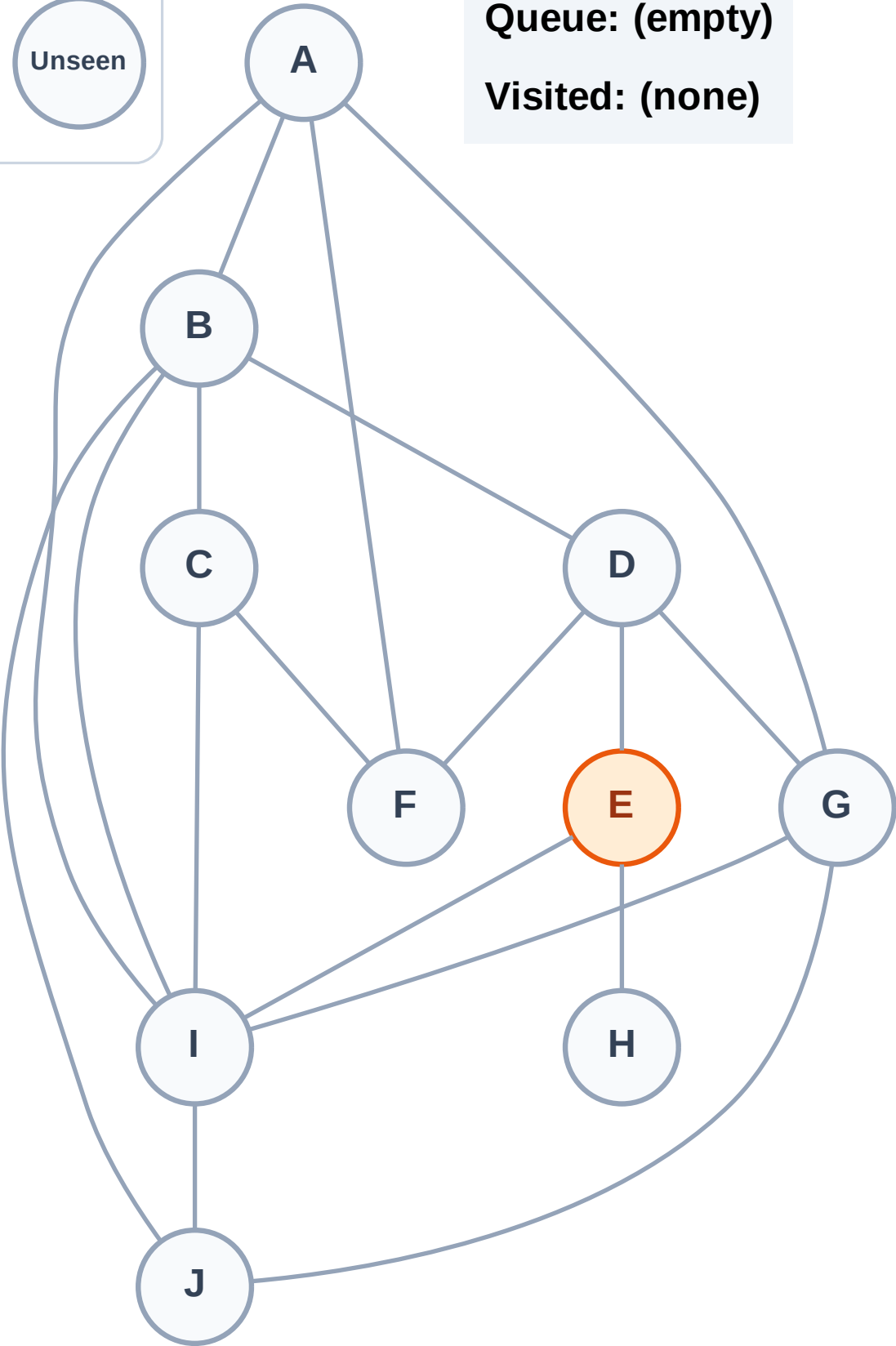
Complete

Processing

Discovered

Unseen

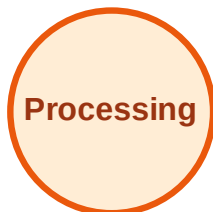
Queue: (empty)
Visited: (none)



BFS Traversal

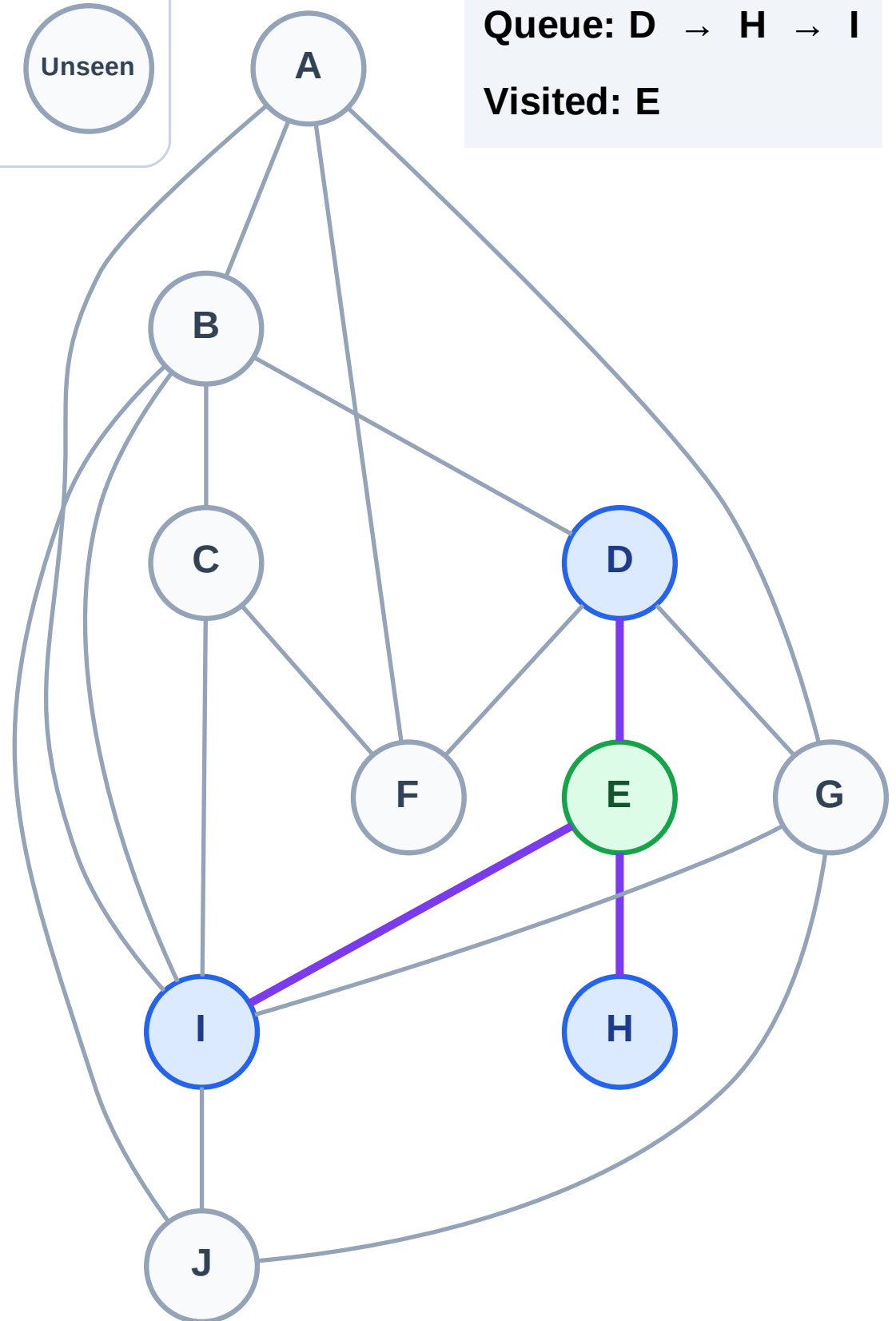
Step 3: Discover D, H, I from E

Legend



Queue: D → H → I

Visited: E



BFS Traversal

Step 4: Process node D

Legend

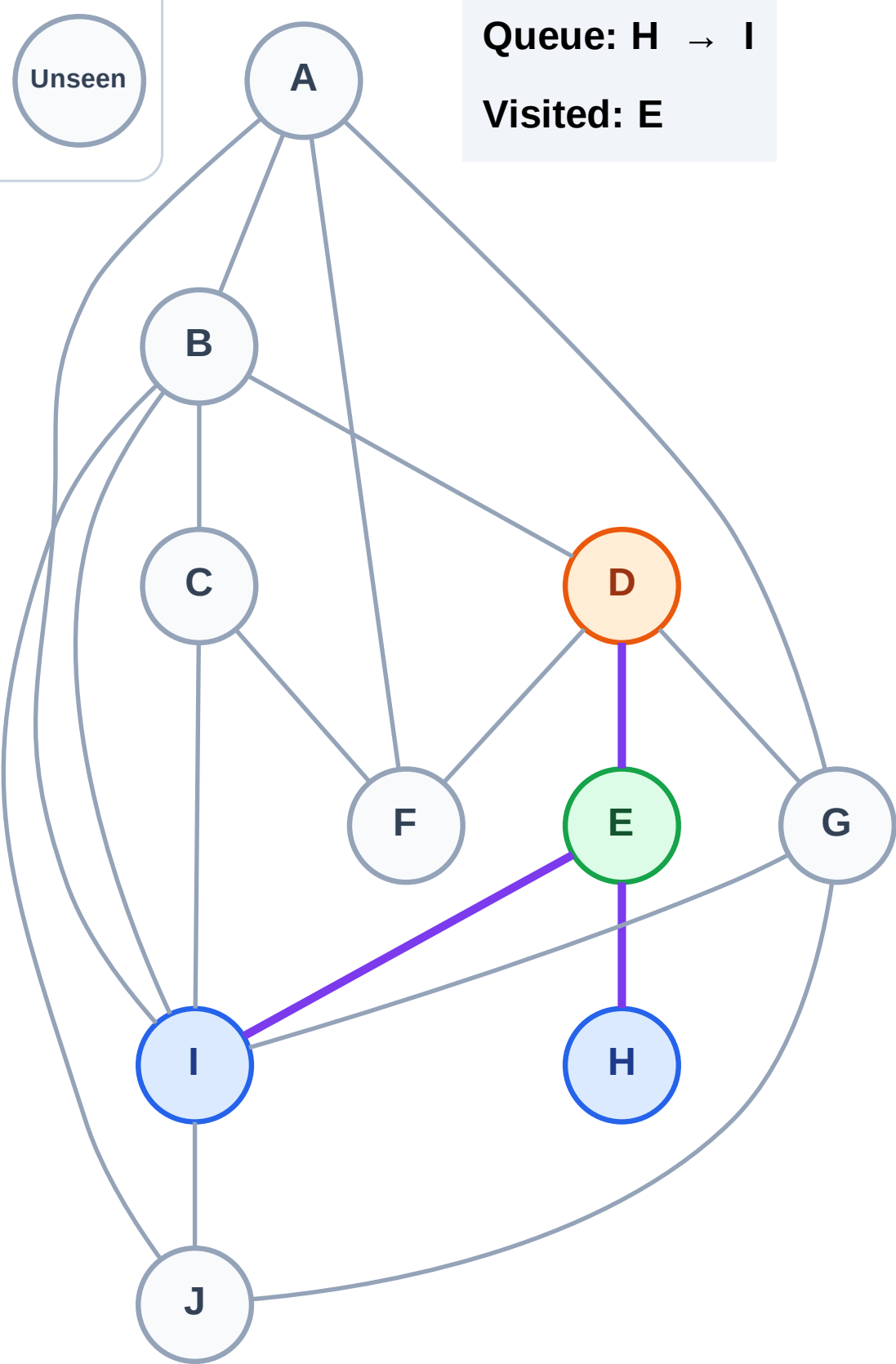
Complete

Processing

Discovered

Unseen

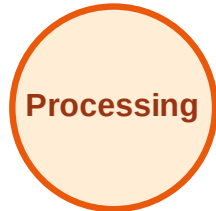
Queue: H → I
Visited: E



BFS Traversal

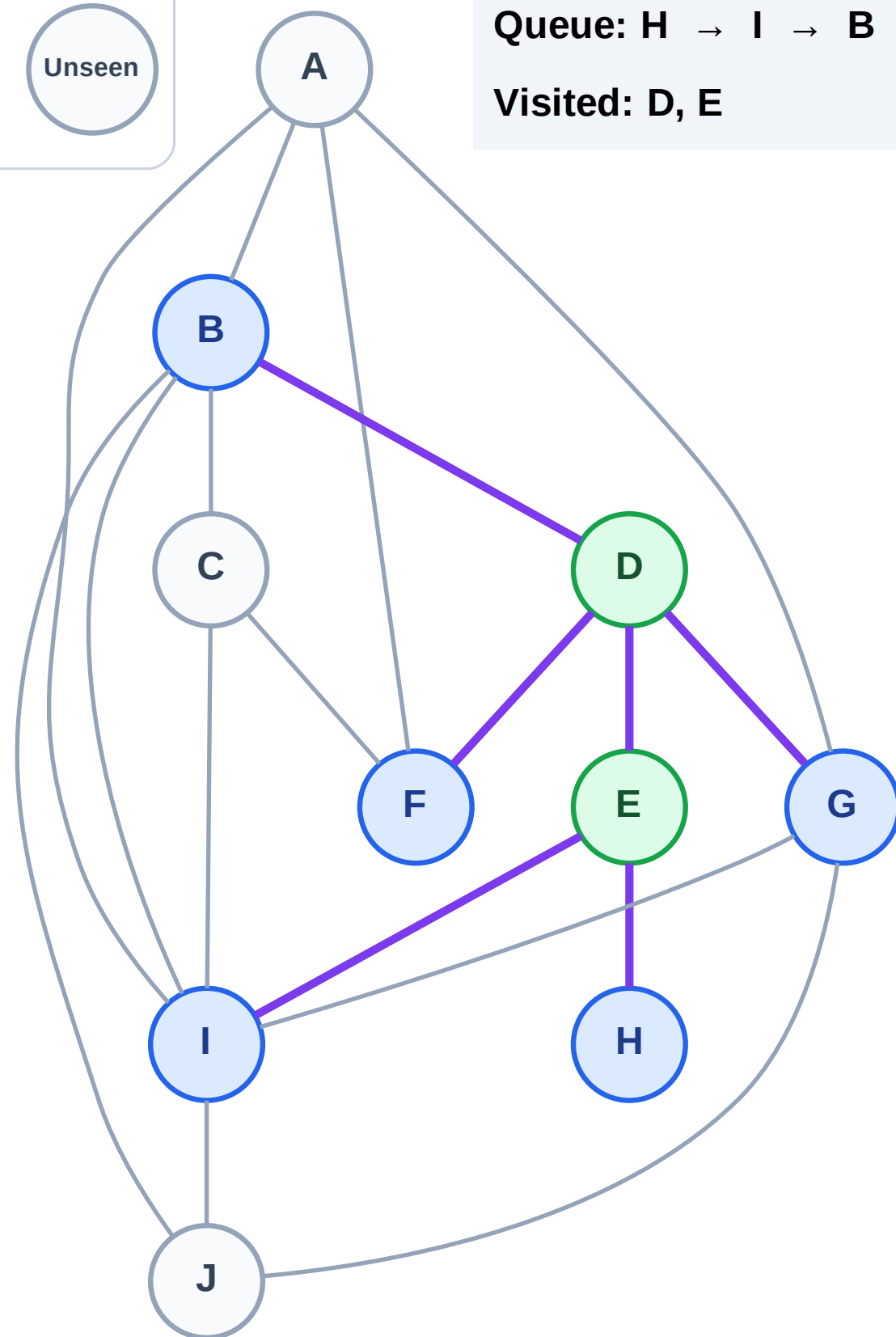
Step 5: Discover B, F, G from D

Legend



Queue: H → I → B → F → G

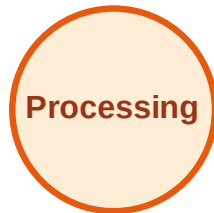
Visited: D, E



BFS Traversal

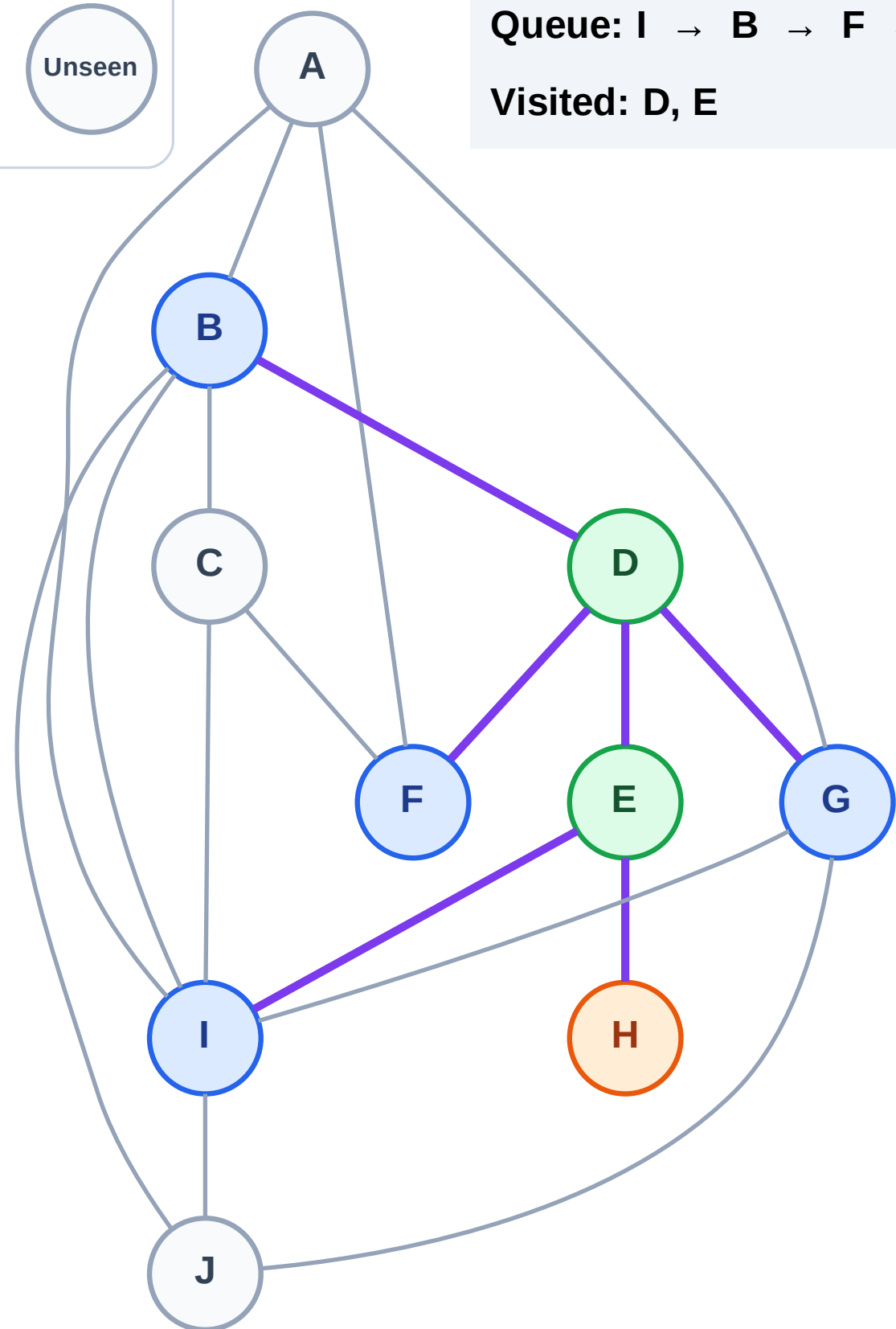
Step 6: Process node H

Legend



Queue: I → B → F → G

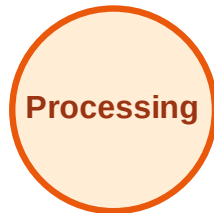
Visited: D, E



BFS Traversal

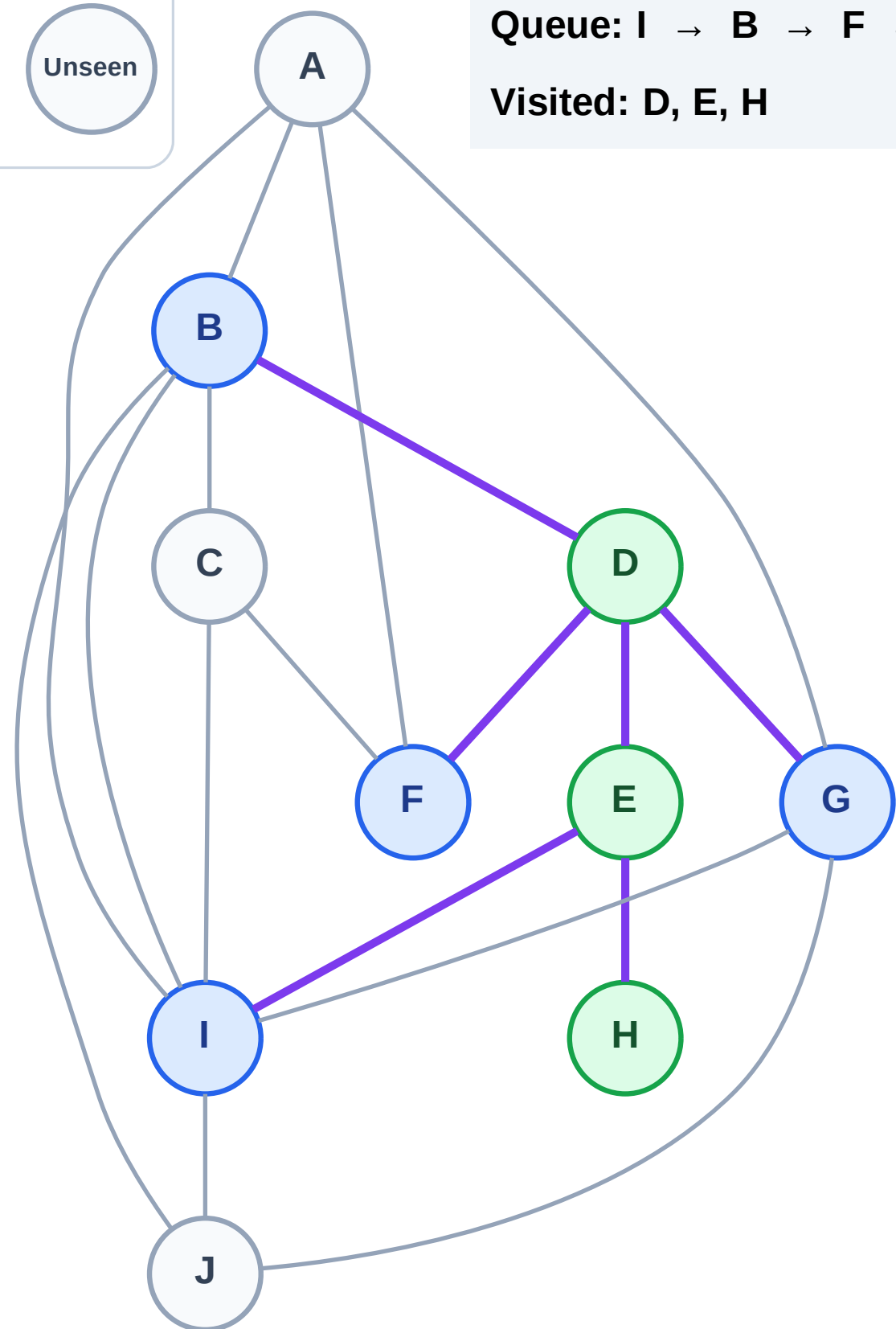
Step 7: No new neighbors from H

Legend



Queue: I → B → F → G

Visited: D, E, H



BFS Traversal

Step 8: Process node I

Legend

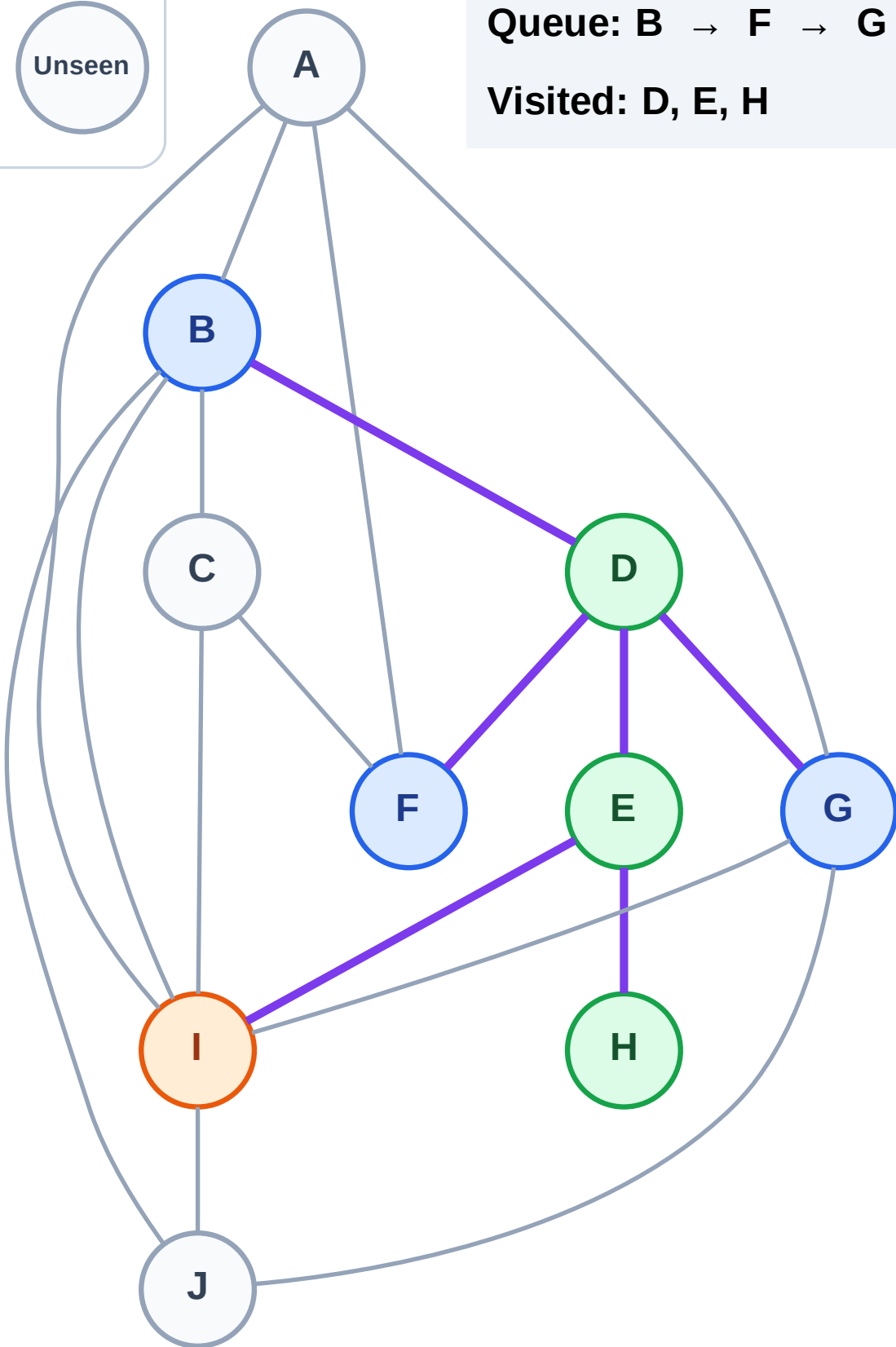
Complete

Processing

Discovered

Unseen

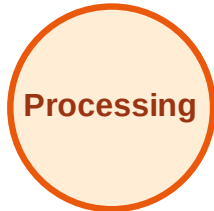
Queue: B → F → G
Visited: D, E, H



BFS Traversal

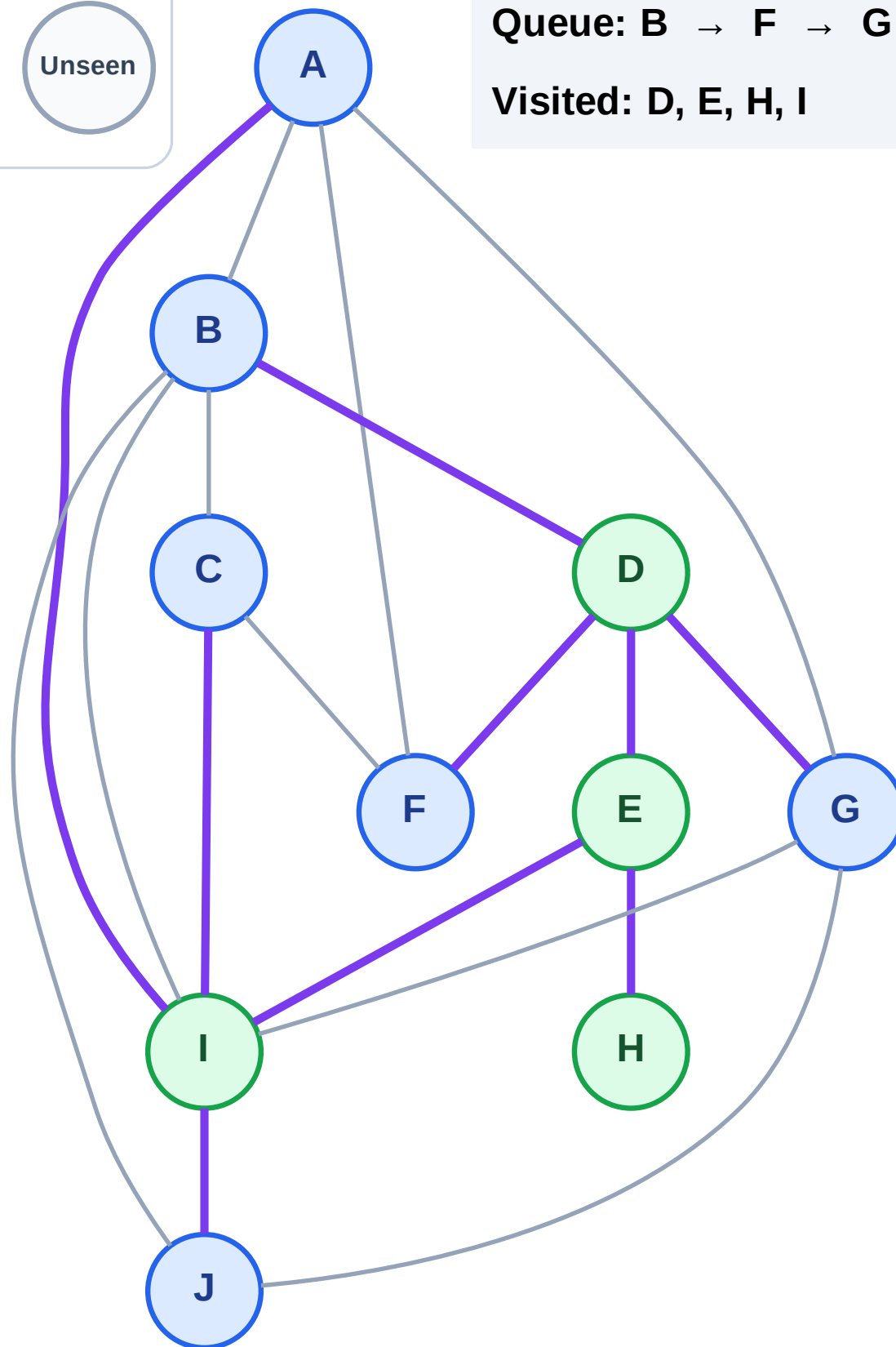
Step 9: Discover A, C, J from I

Legend



Queue: B → F → G → A → C → J

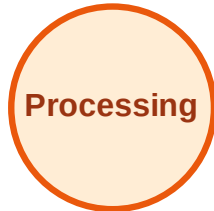
Visited: D, E, H, I



BFS Traversal

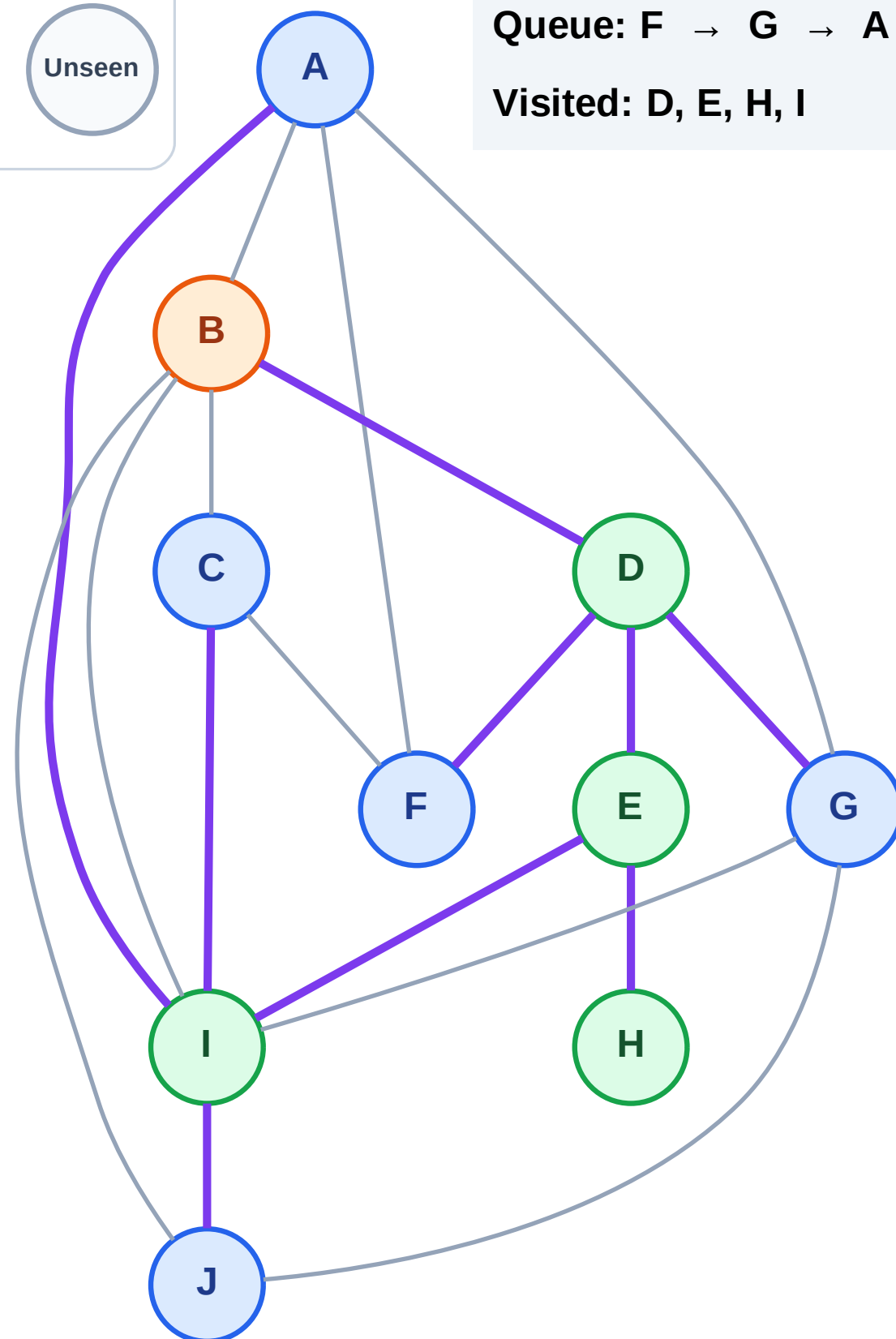
Step 10: Process node B

Legend



Queue: F → G → A → C → J

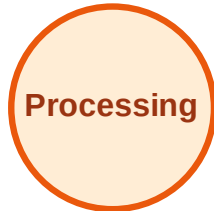
Visited: D, E, H, I



BFS Traversal

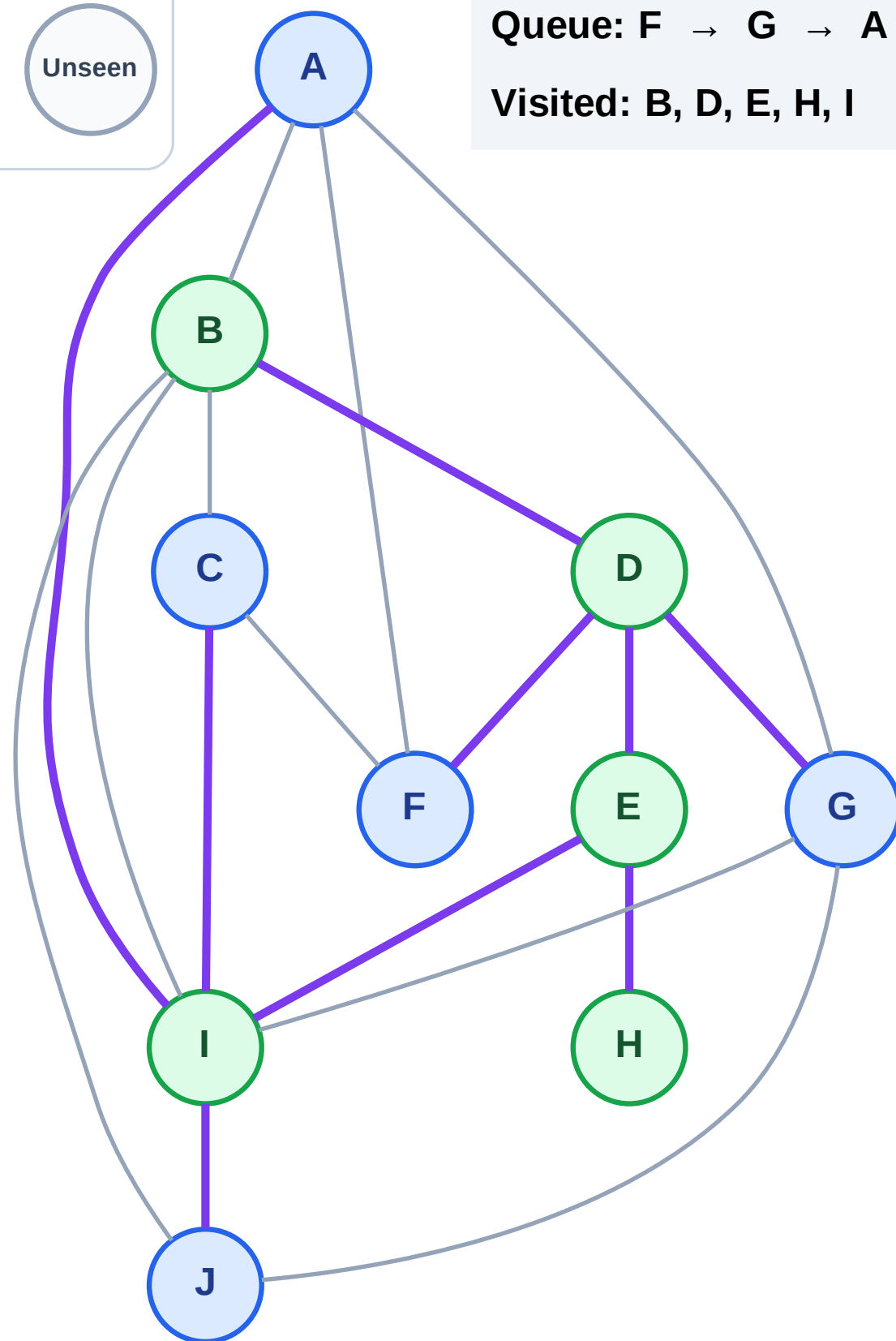
Step 11: No new neighbors from B

Legend



Queue: F → G → A → C → J

Visited: B, D, E, H, I



BFS Traversal

Step 12: Process node F

Legend

Complete

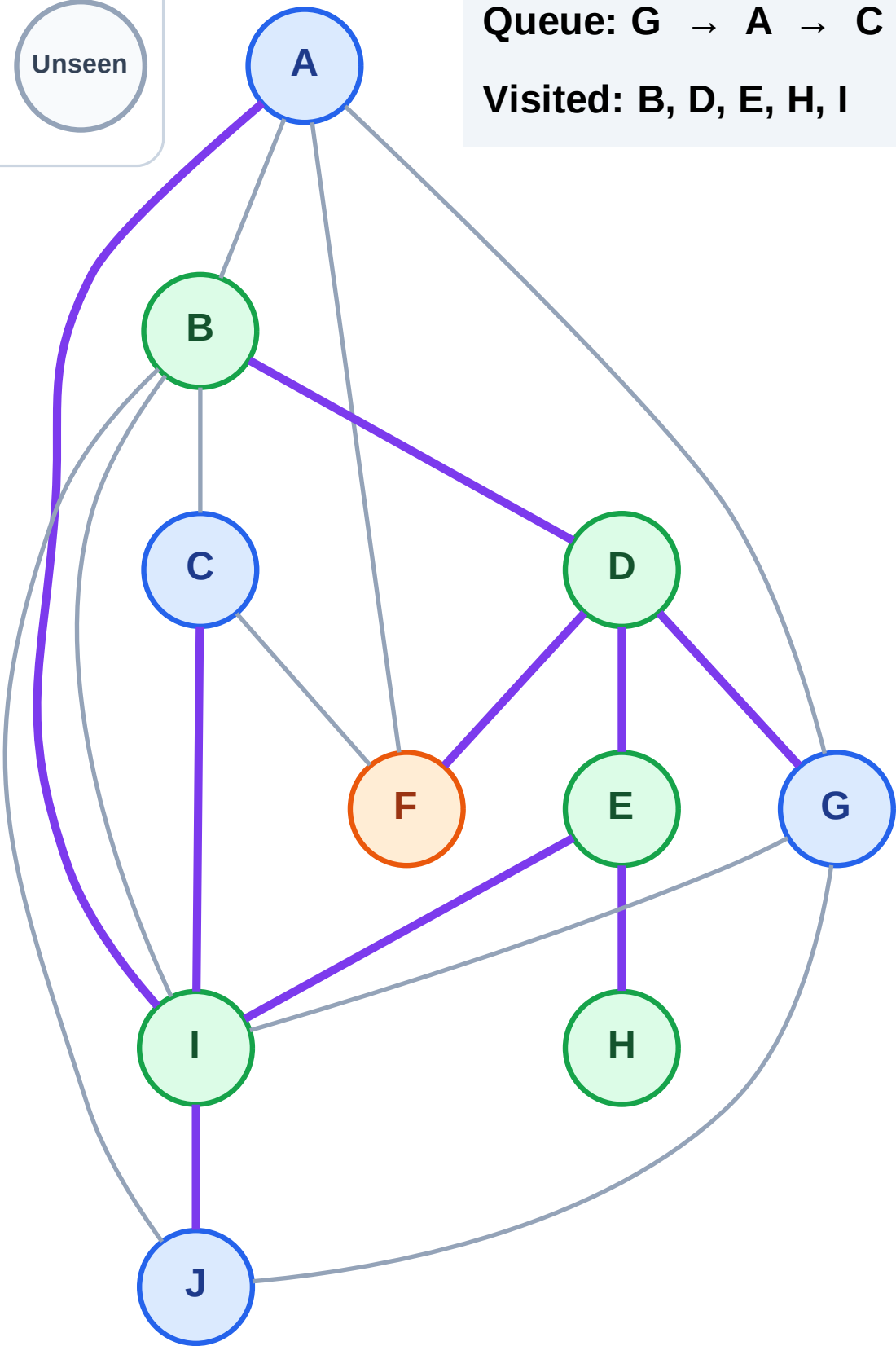
Processing

Discovered

Unseen

Queue: G → A → C → J

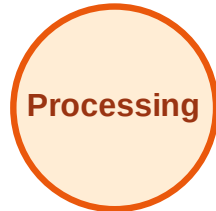
Visited: B, D, E, H, I



BFS Traversal

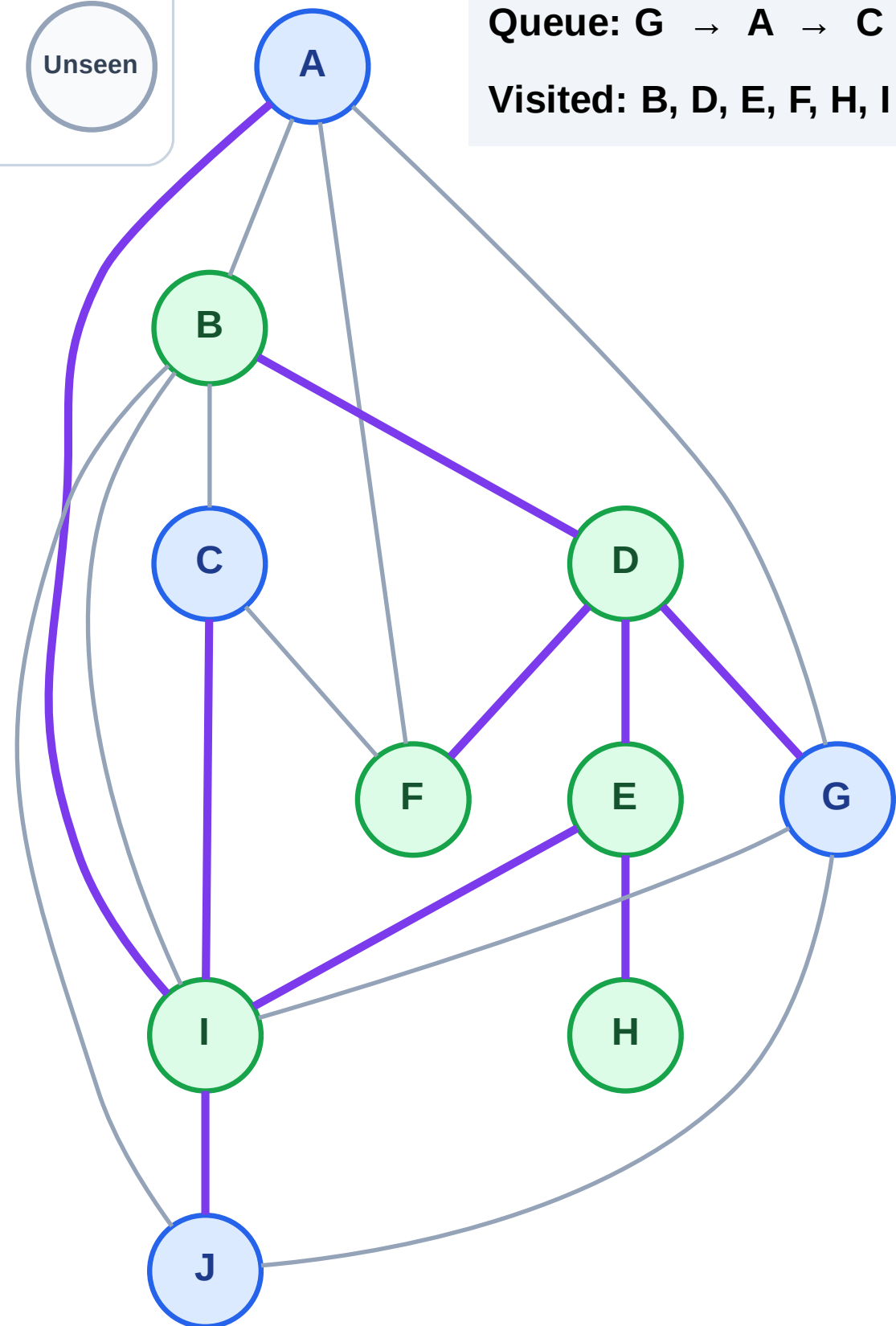
Step 13: No new neighbors from F

Legend



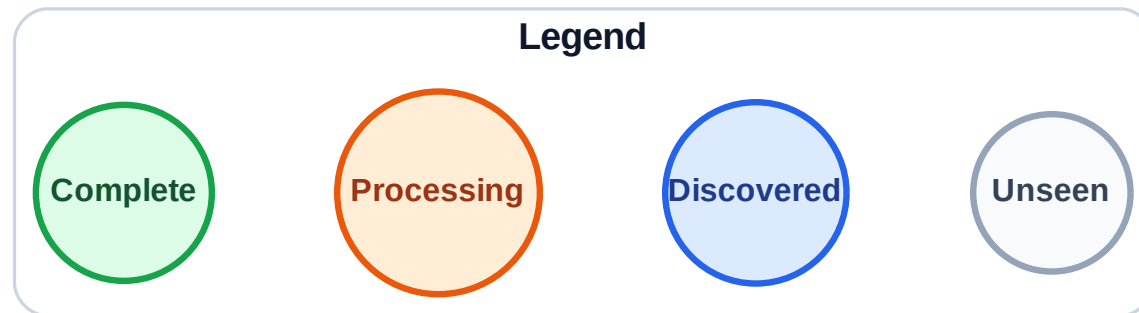
Queue: G → A → C → J

Visited: B, D, E, F, H, I

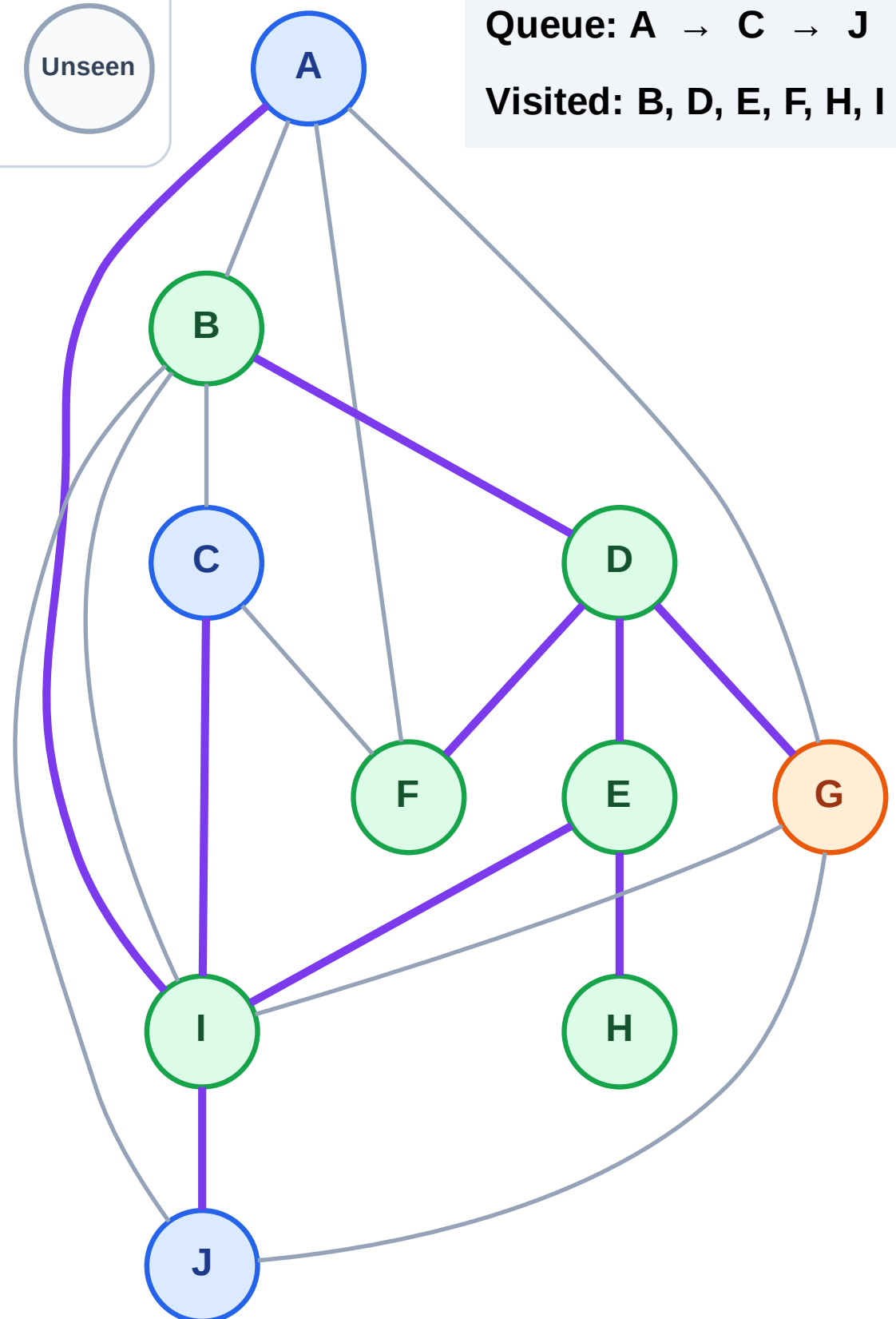


BFS Traversal

Step 14: Process node G



Queue: A → C → J
Visited: B, D, E, F, H, I



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

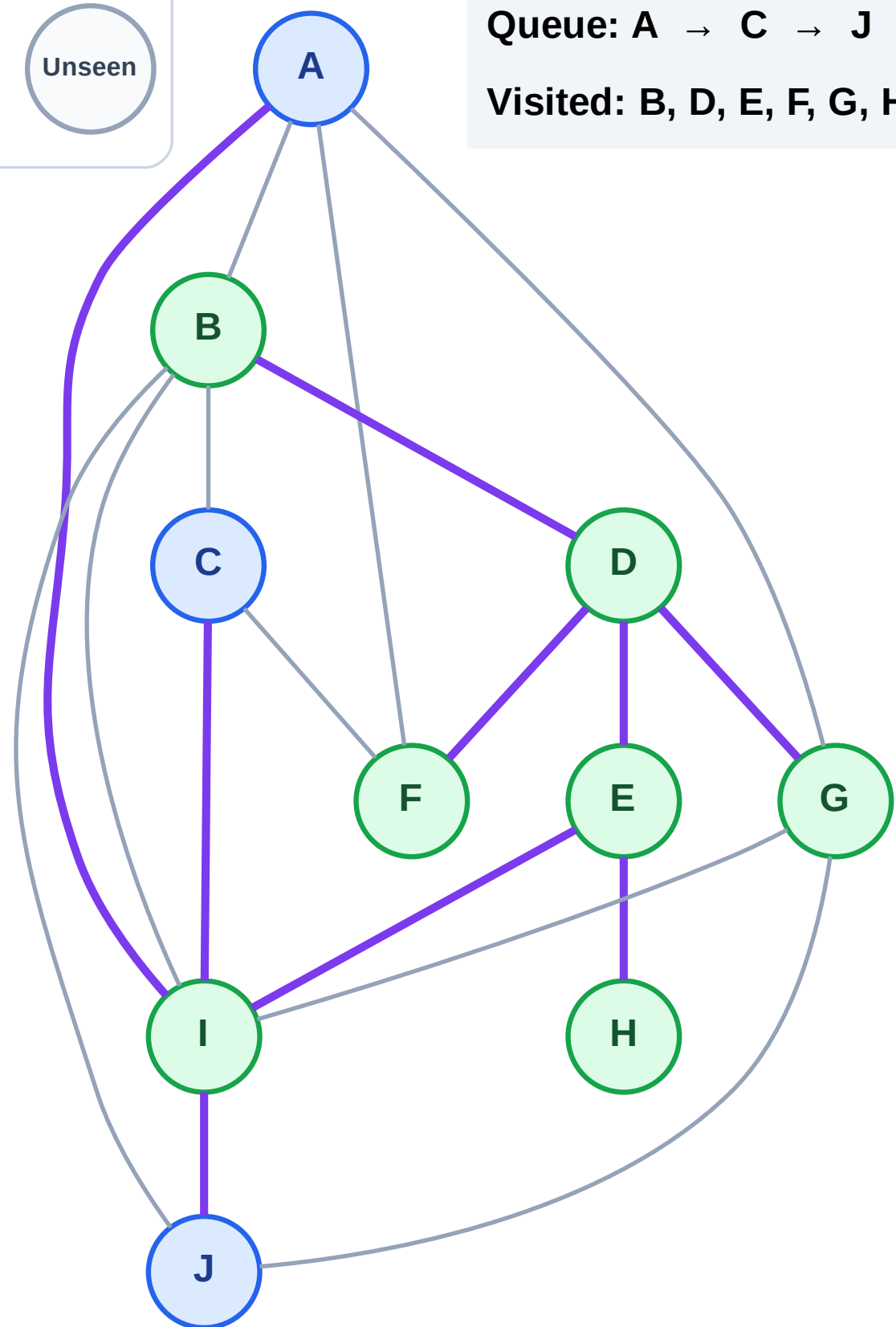
Processing

Discovered

Unseen

Queue: A → C → J

Visited: B, D, E, F, G, H, I



BFS Traversal

Step 16: Process node A

Legend

Complete

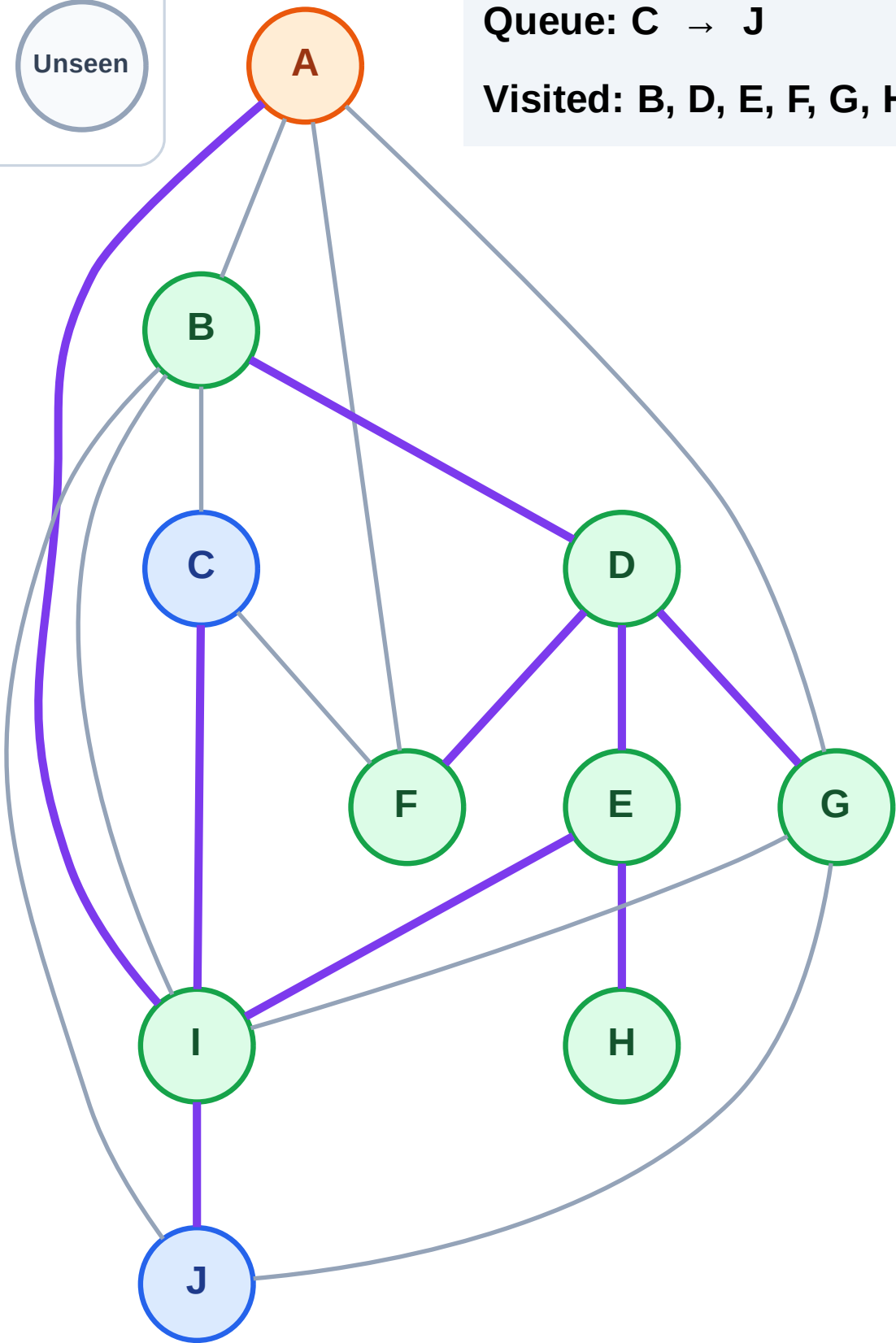
Processing

Discovered

Unseen

Queue: C → J

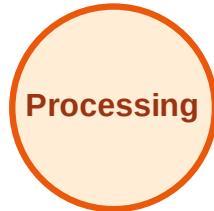
Visited: B, D, E, F, G, H, I



BFS Traversal

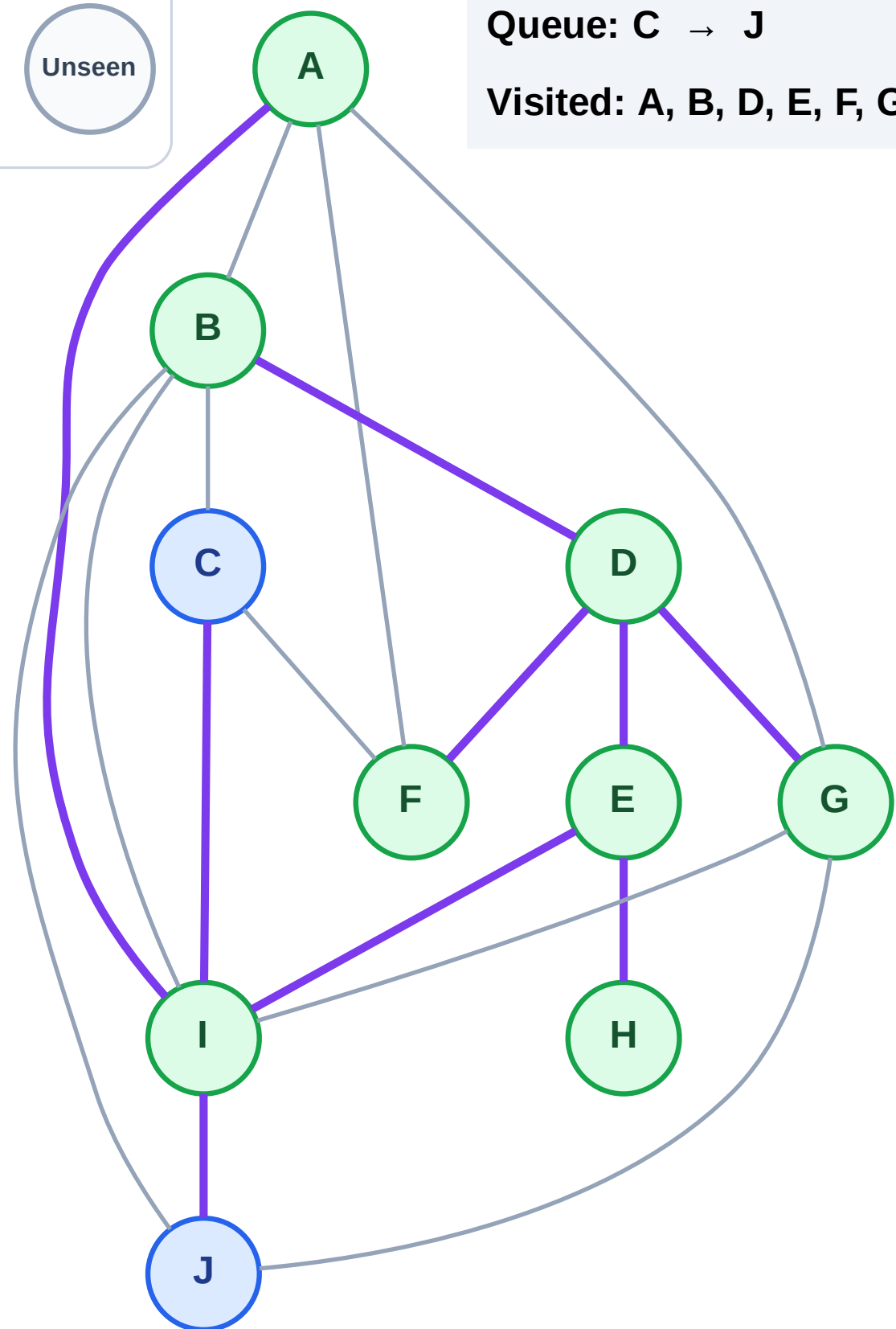
Step 17: No new neighbors from A

Legend



Queue: C → J

Visited: A, B, D, E, F, G, H, I



BFS Traversal

Step 18: Process node C

Legend

Complete

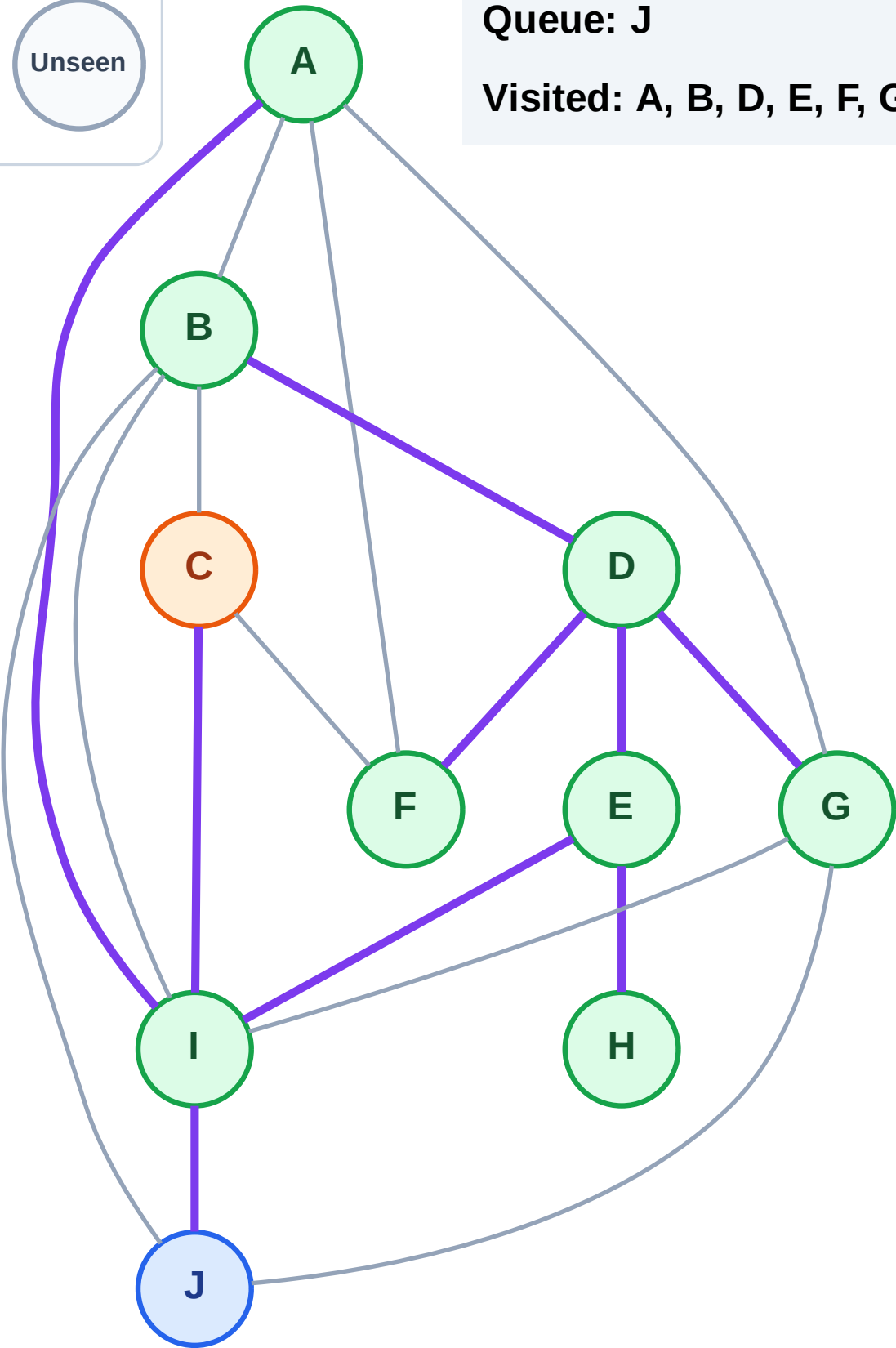
Processing

Discovered

Unseen

Queue: J

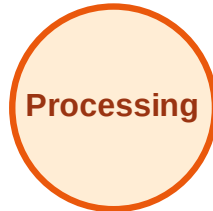
Visited: A, B, D, E, F, G, H, I



BFS Traversal

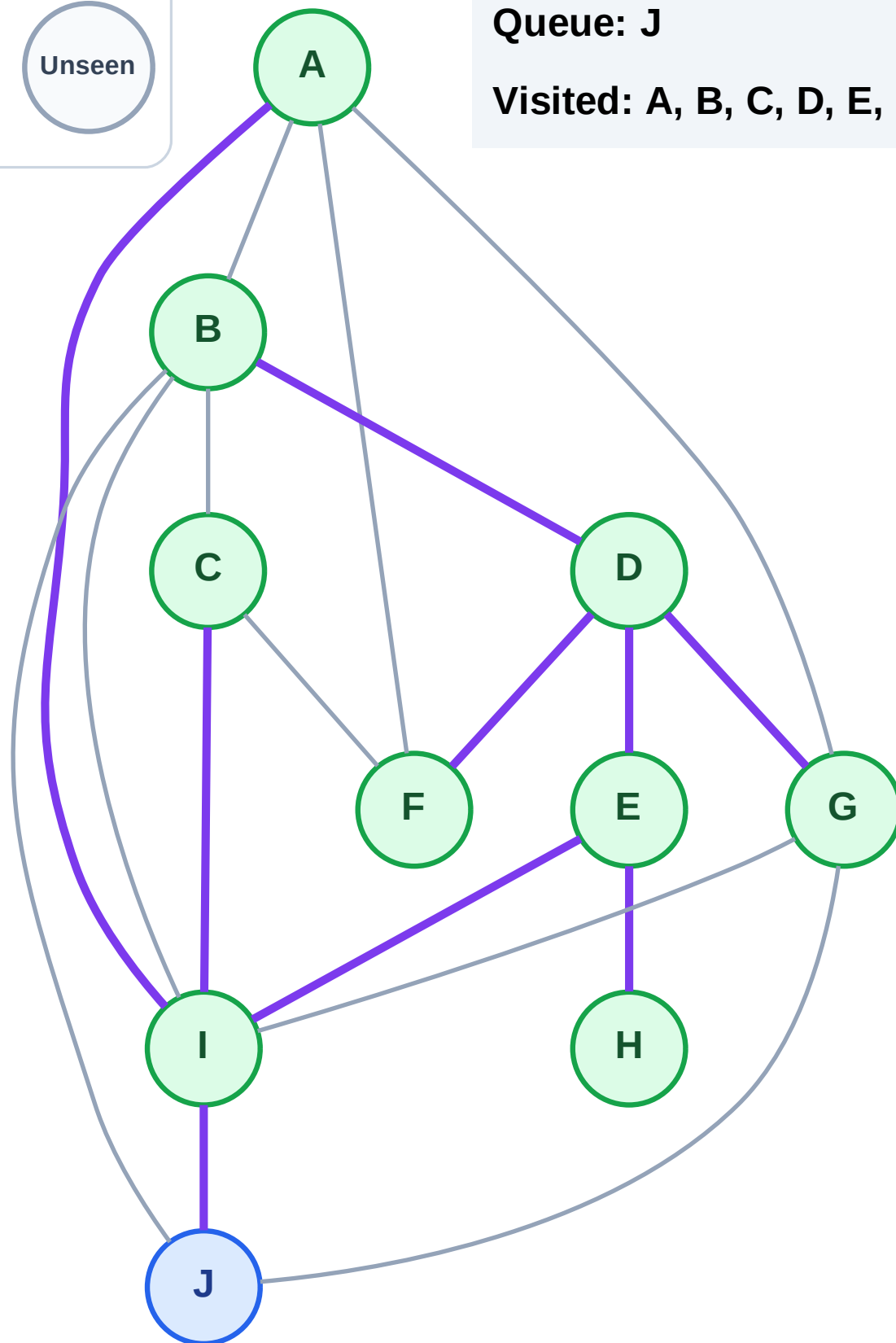
Step 19: No new neighbors from C

Legend



Queue: J

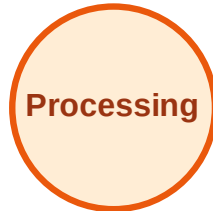
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

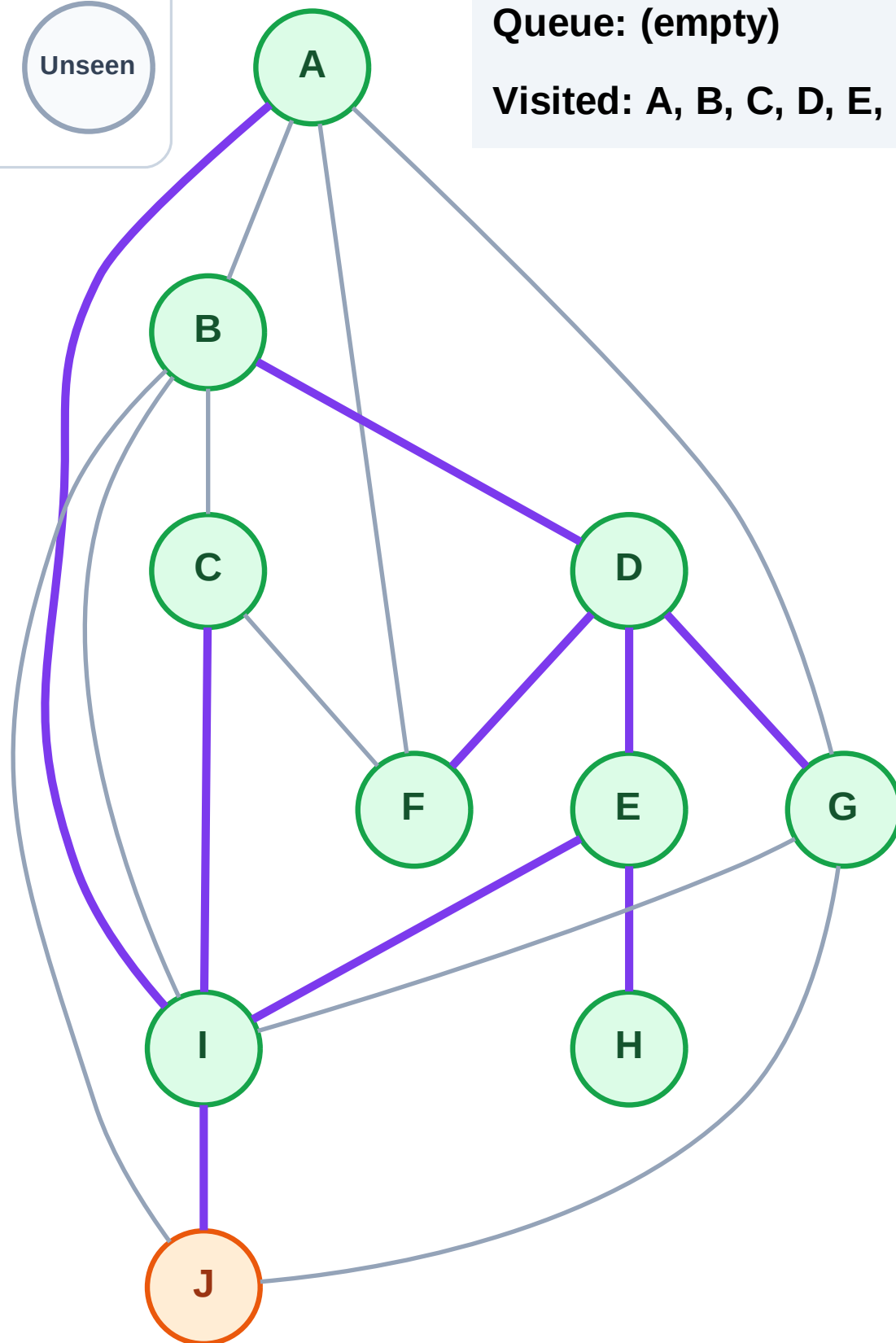
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

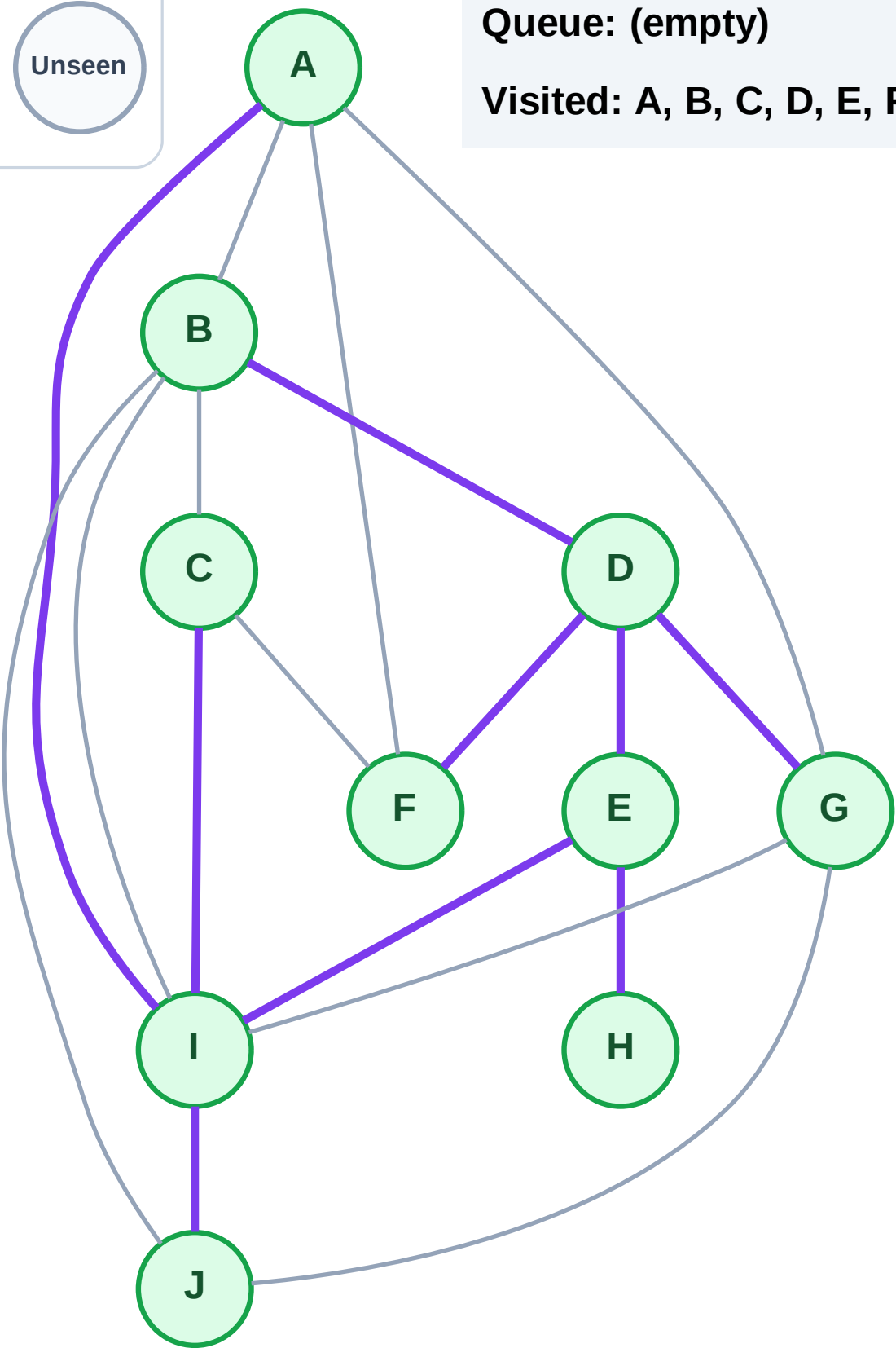
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

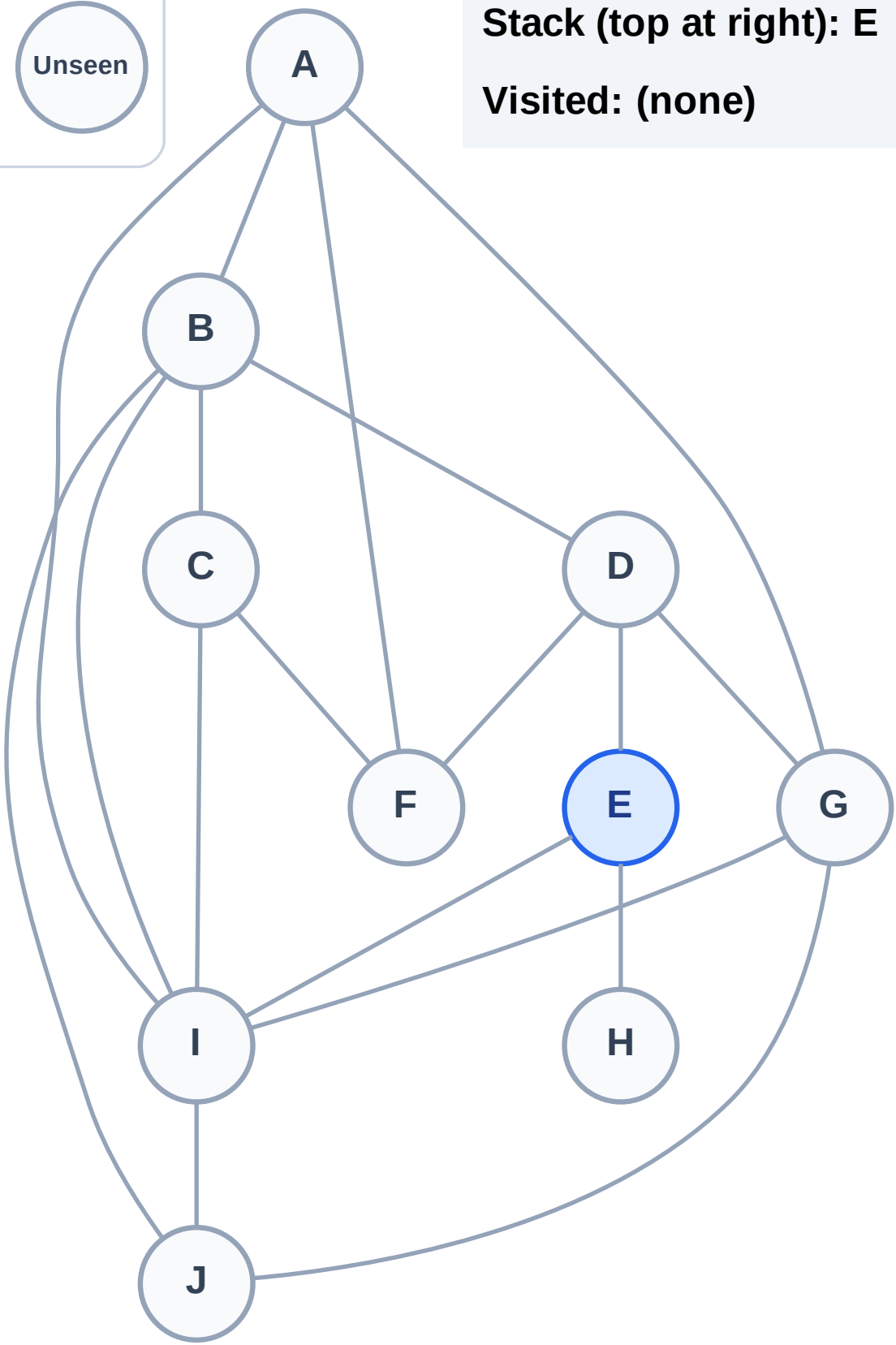
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

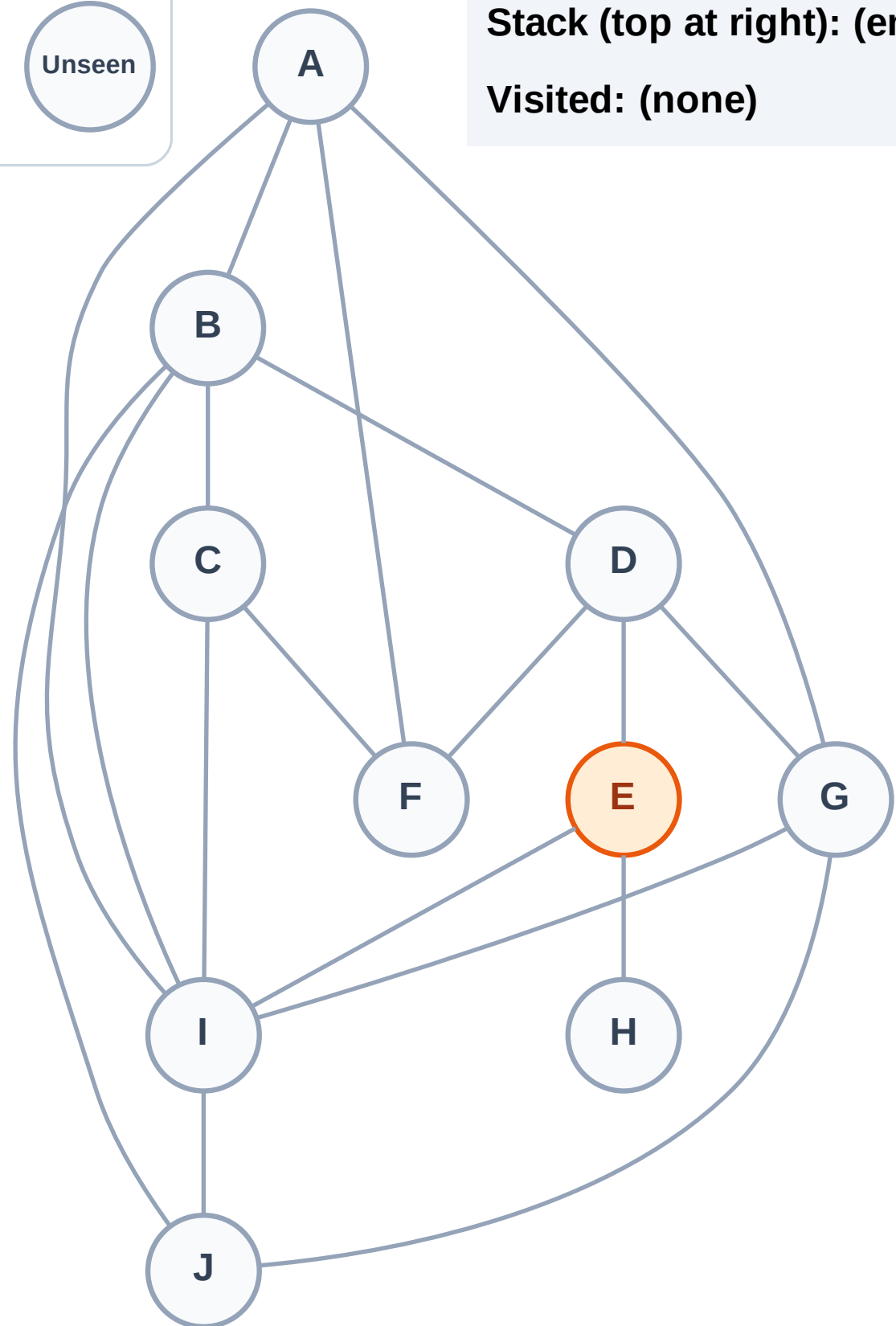
Processing

Discovered

Unseen

Stack (top at right): (empty)

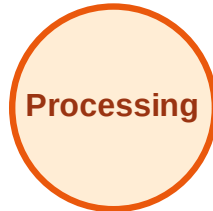
Visited: (none)



DFS Traversal

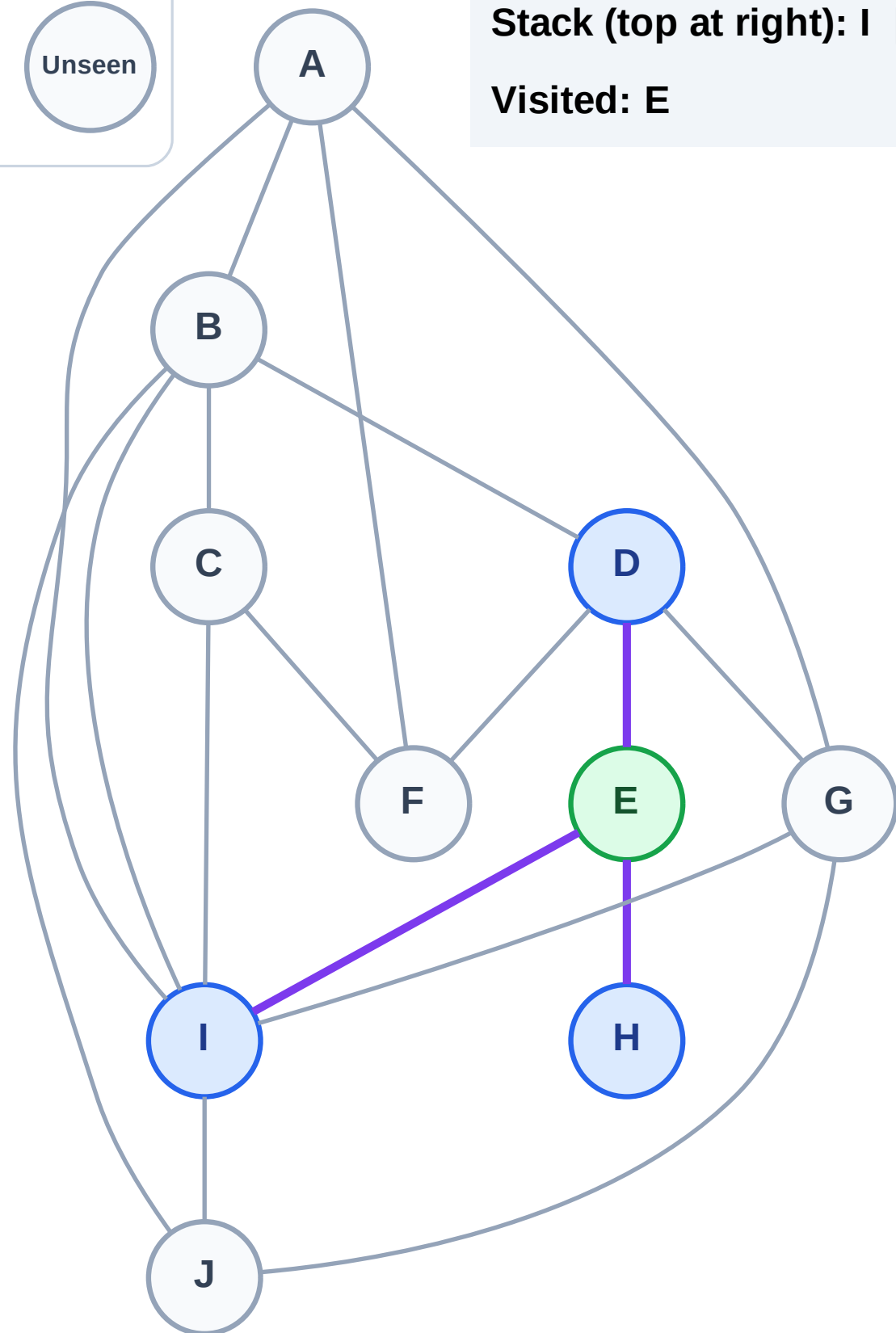
Step 3: Discover I, H, D from E

Legend



Stack (top at right): I | H | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

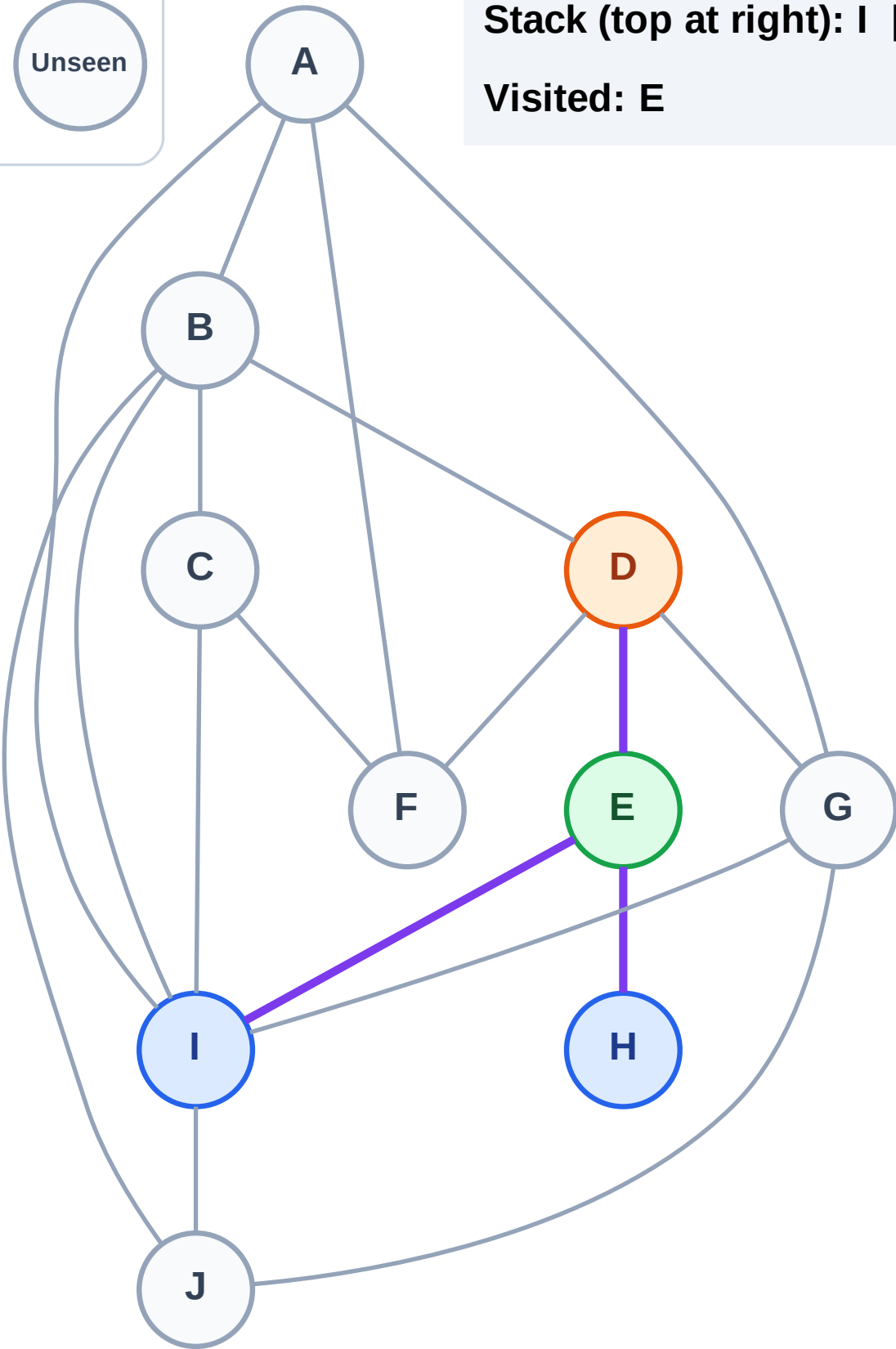
Complete

Processing

Discovered

Unseen

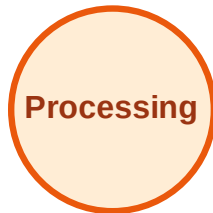
Stack (top at right): I | H
Visited: E



DFS Traversal

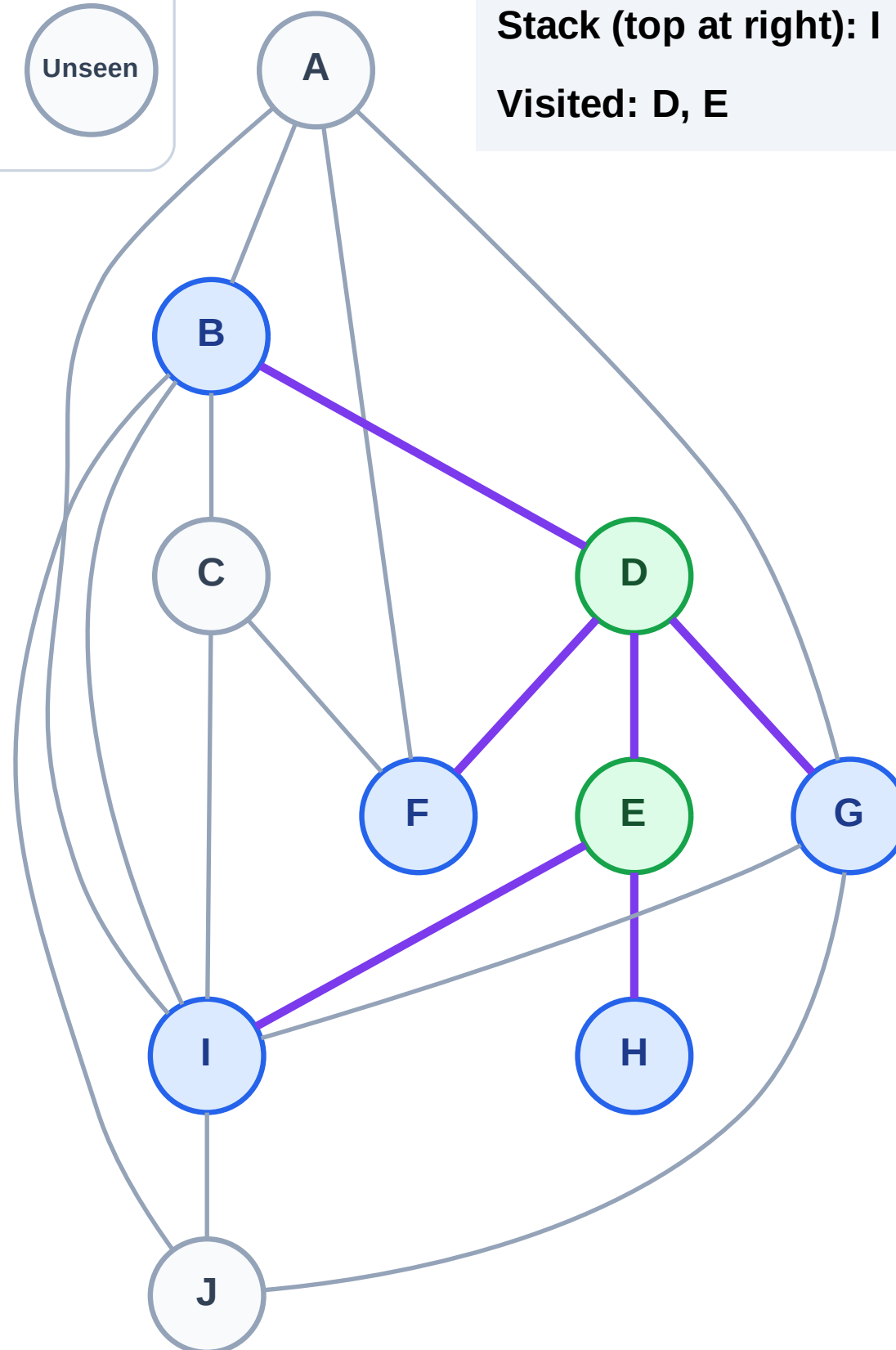
Step 5: Discover G, F, B from D

Legend



Stack (top at right): I | H | G | F | B

Visited: D, E



DFS Traversal

Step 6: Process node B

Legend

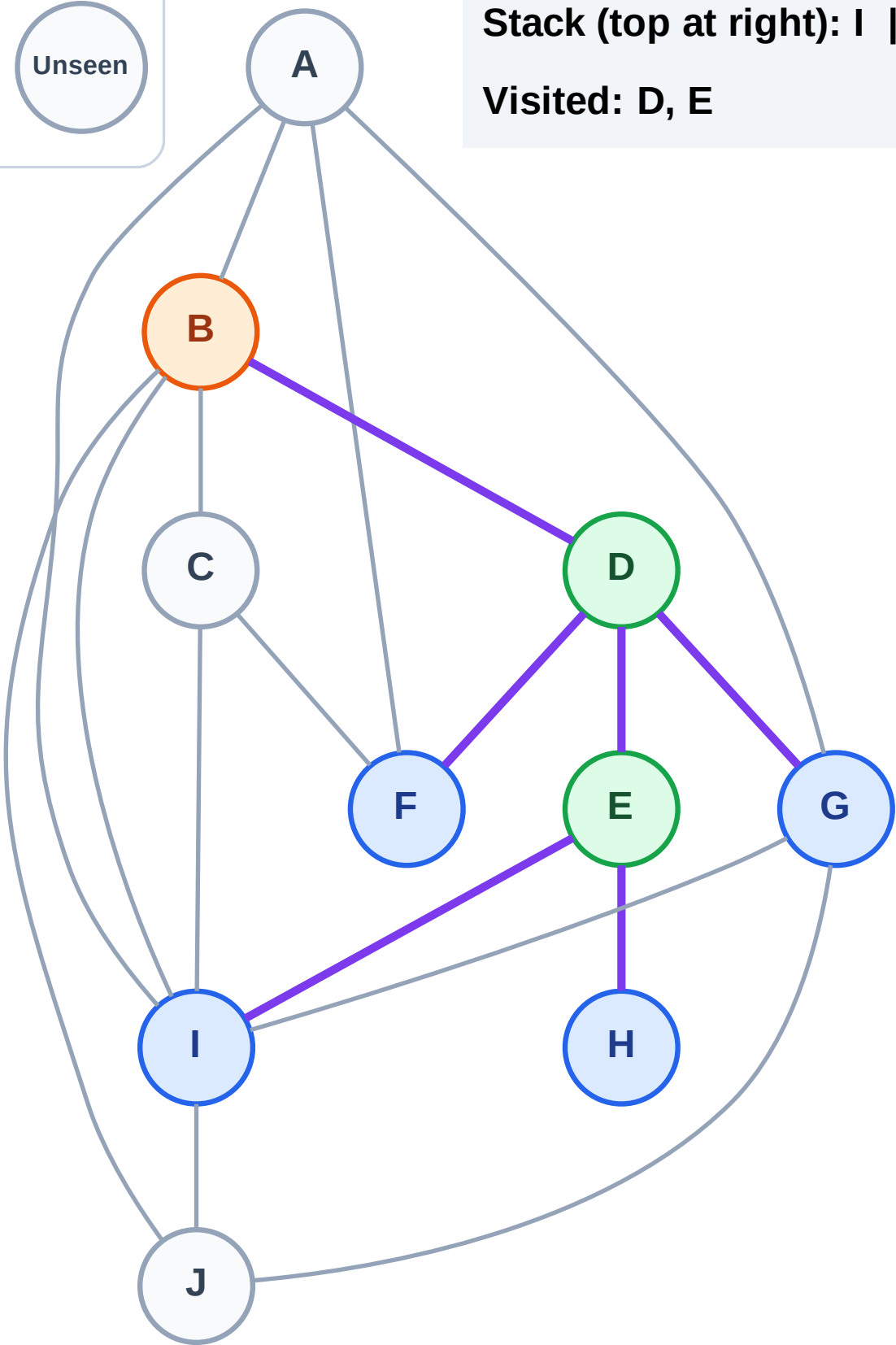
Complete

Processing

Discovered

Unseen

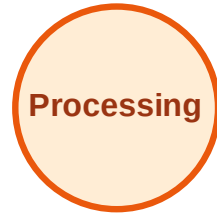
Stack (top at right): I | H | G | F
Visited: D, E



DFS Traversal

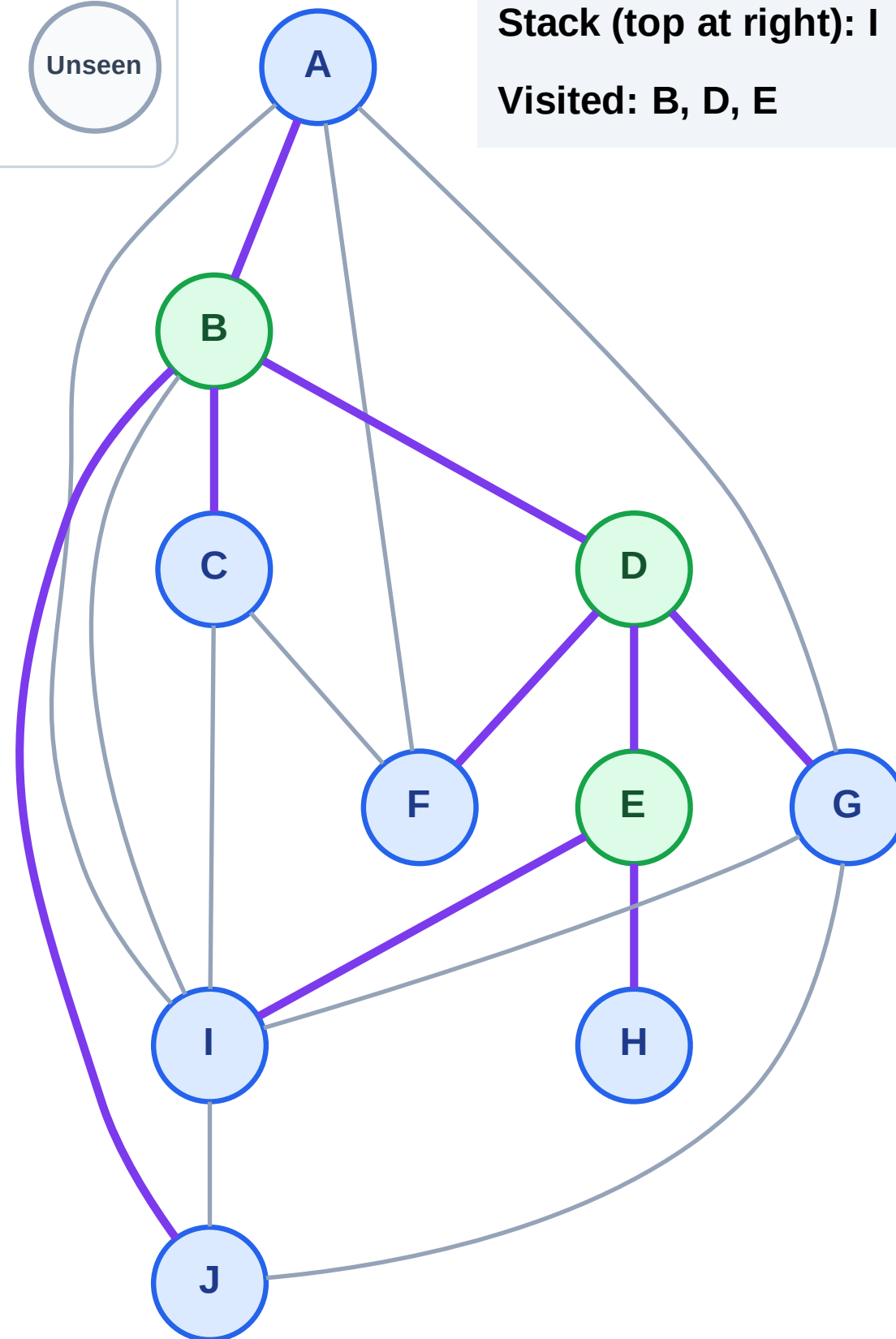
Step 7: Discover J, C, A from B

Legend



Stack (top at right): I | H | G | F | J | C | A

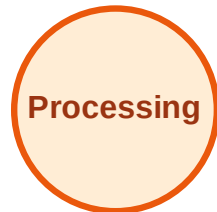
Visited: B, D, E



DFS Traversal

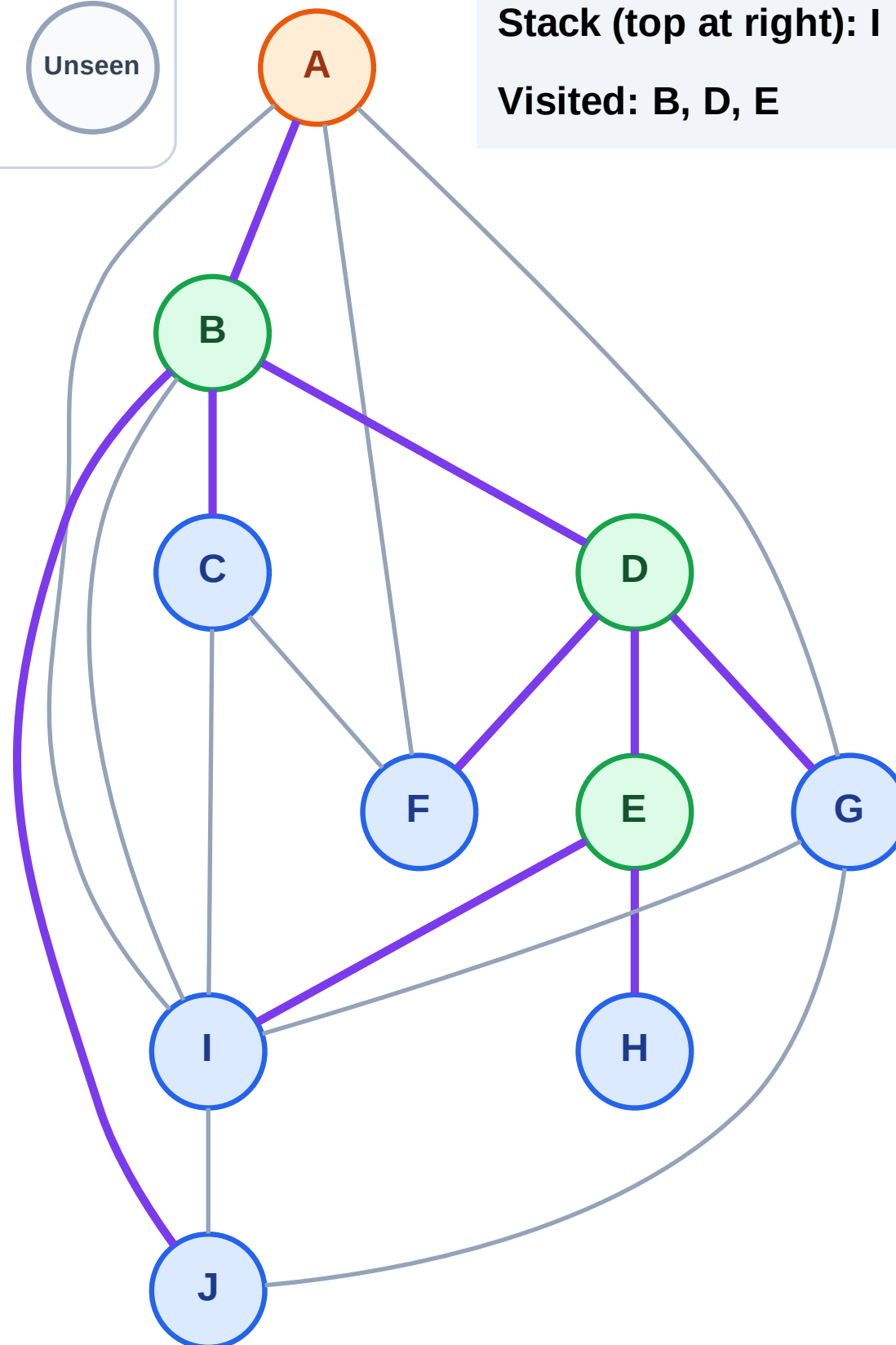
Step 8: Process node A

Legend



Stack (top at right): I | H | G | F | J | C

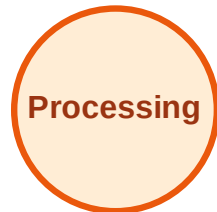
Visited: B, D, E



DFS Traversal

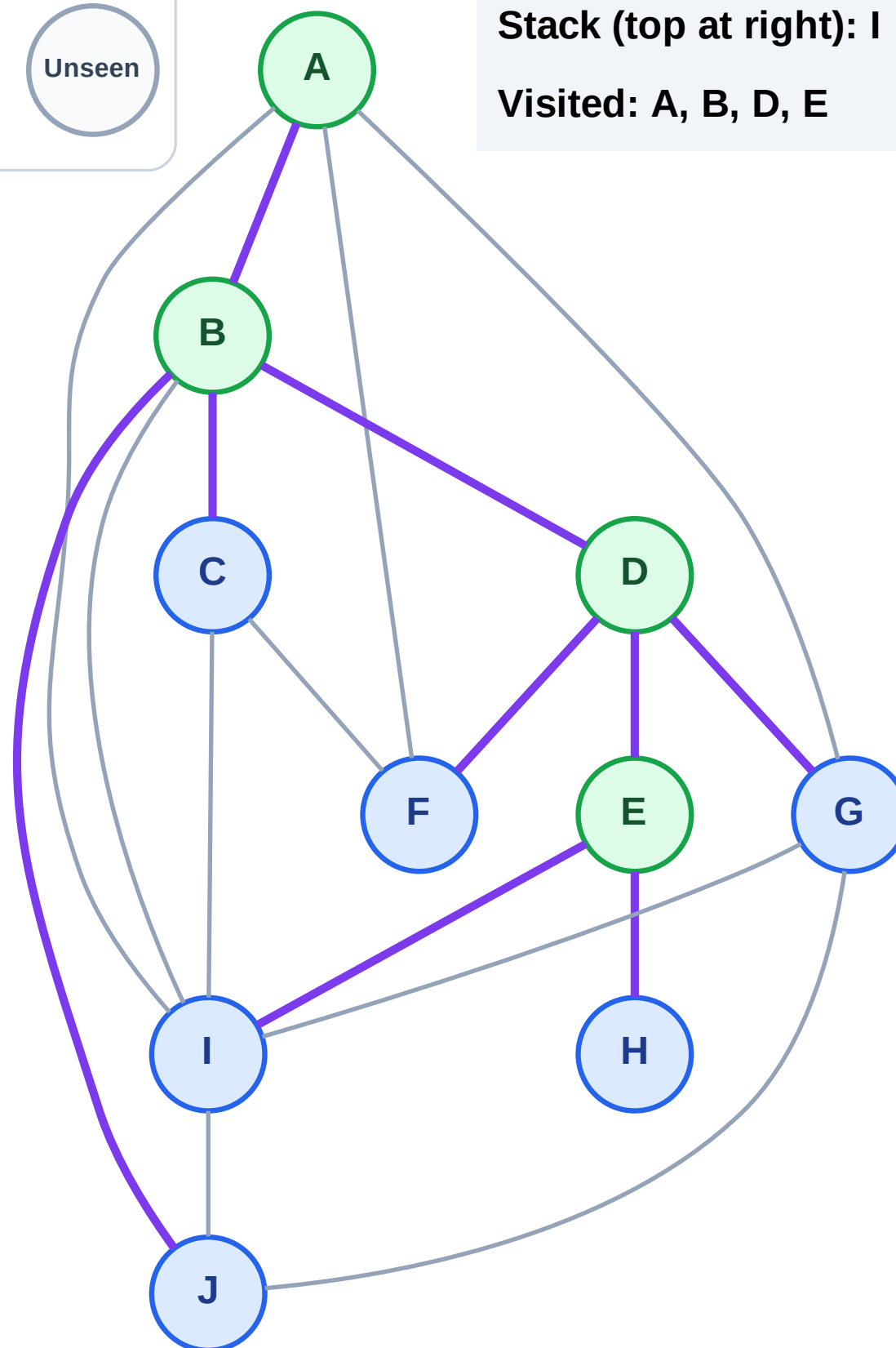
Step 9: No new neighbors from A

Legend



Stack (top at right): I | H | G | F | J | C

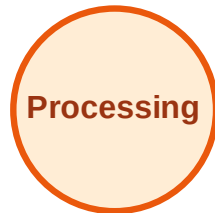
Visited: A, B, D, E



DFS Traversal

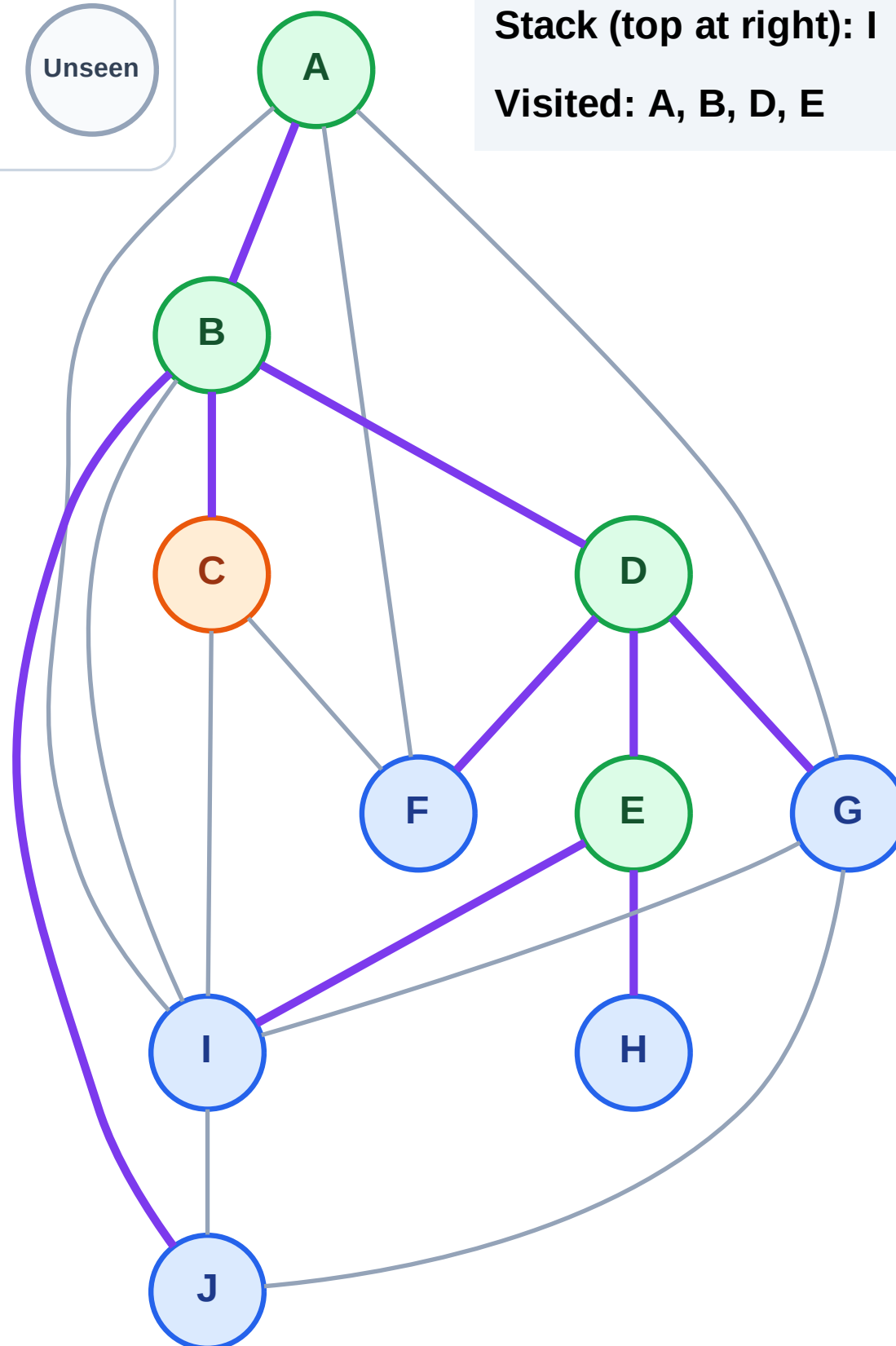
Step 10: Process node C

Legend



Stack (top at right): I | H | G | F | J

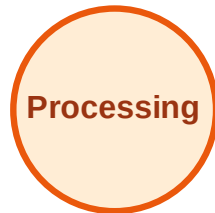
Visited: A, B, D, E



DFS Traversal

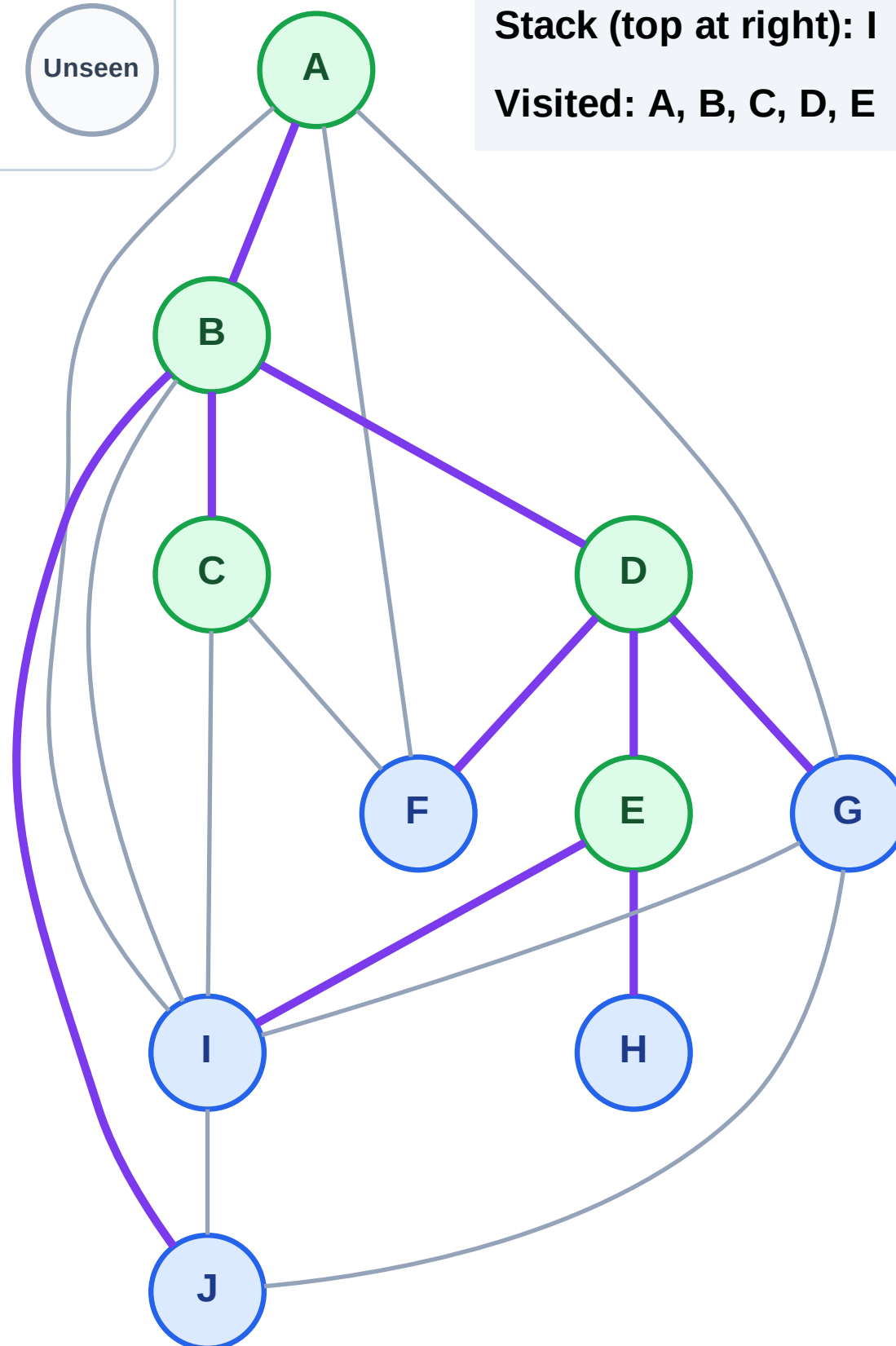
Step 11: No new neighbors from C

Legend



Stack (top at right): I | H | G | F | J

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node J

Legend

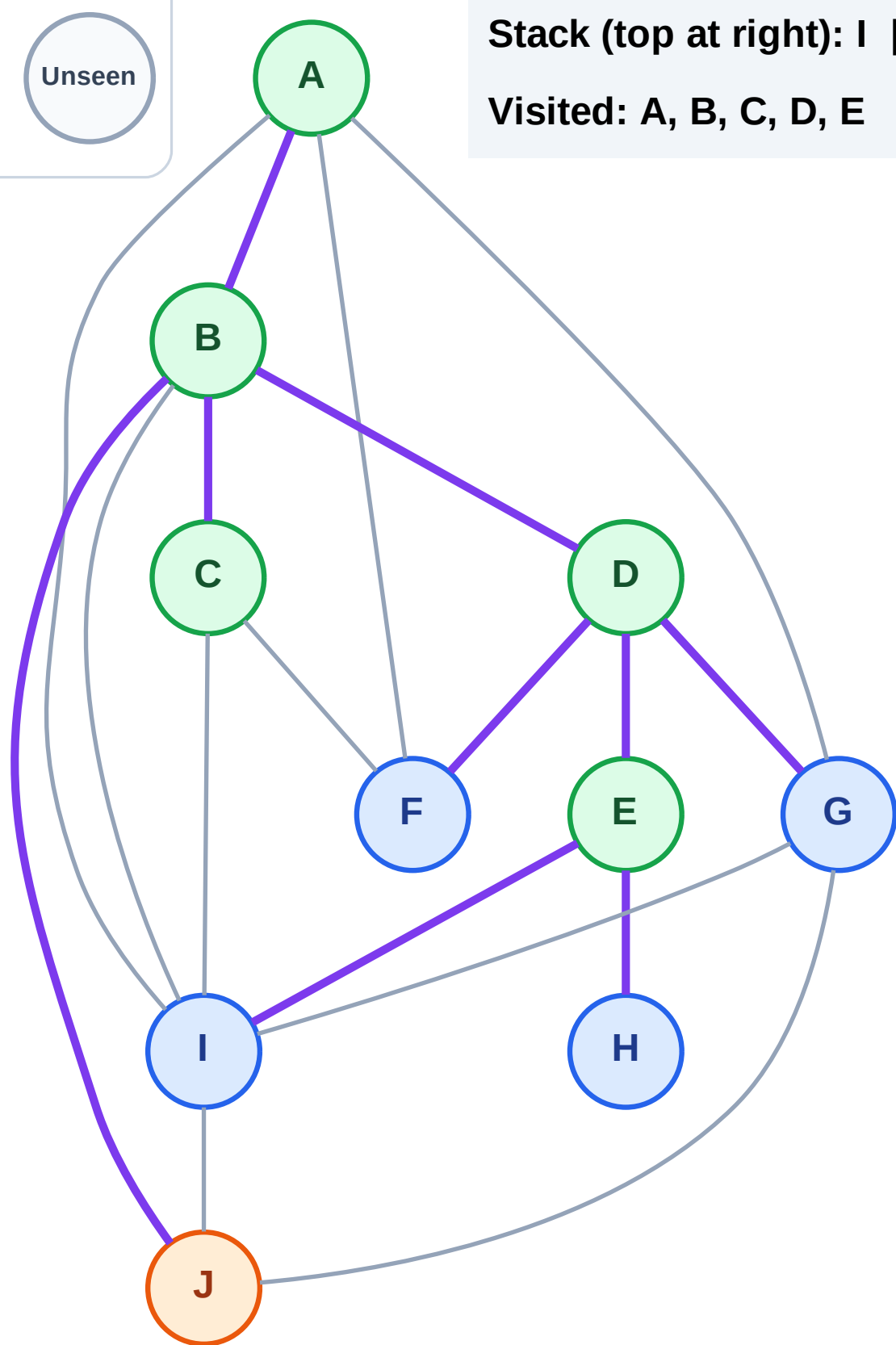
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | G | F
Visited: A, B, C, D, E



DFS Traversal

Step 13: No new neighbors from J

Legend

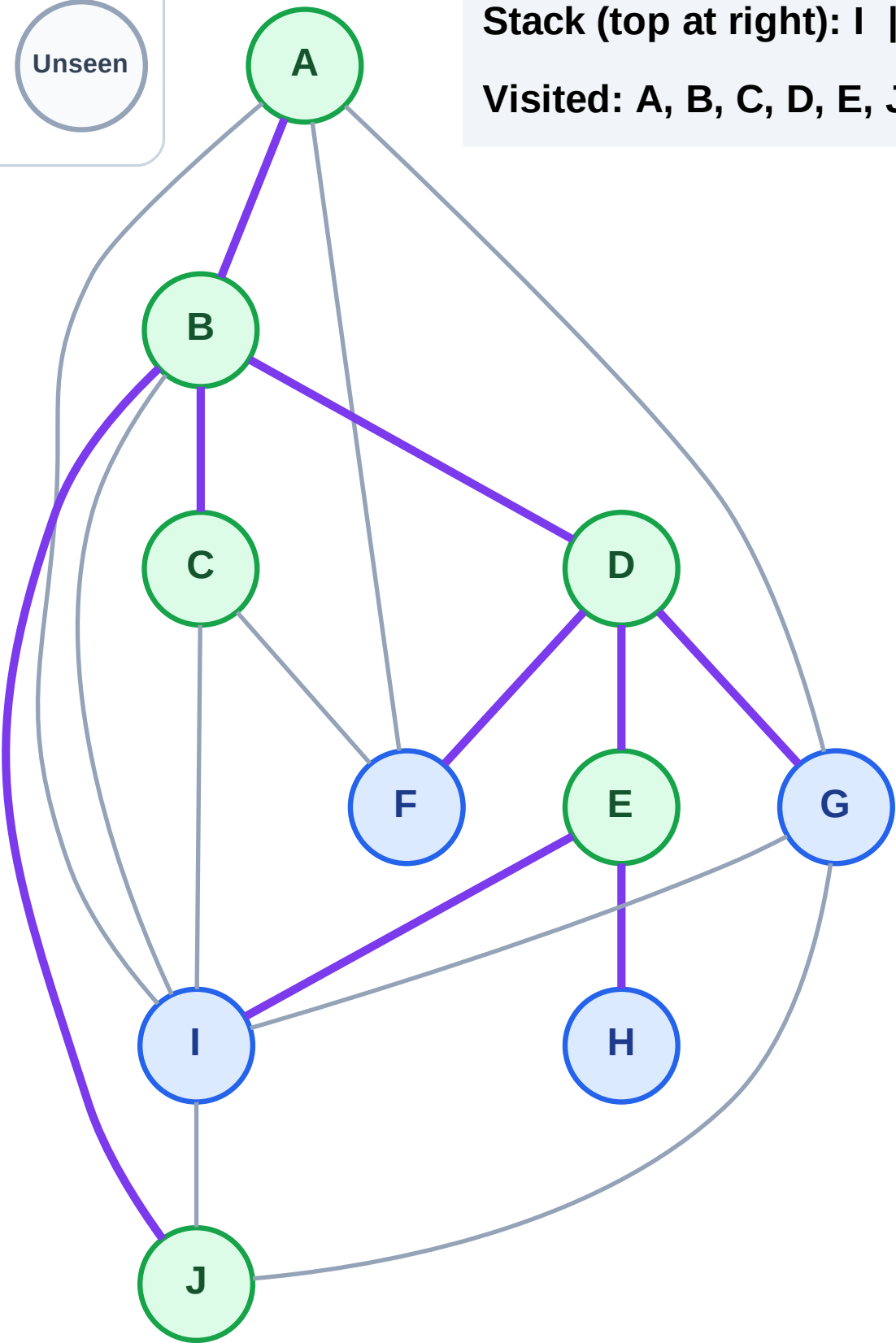
Complete

Processing

Discovered

Unseen

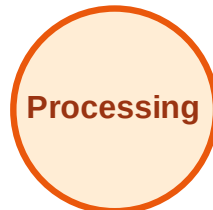
Stack (top at right): I | H | G | F
Visited: A, B, C, D, E, J



DFS Traversal

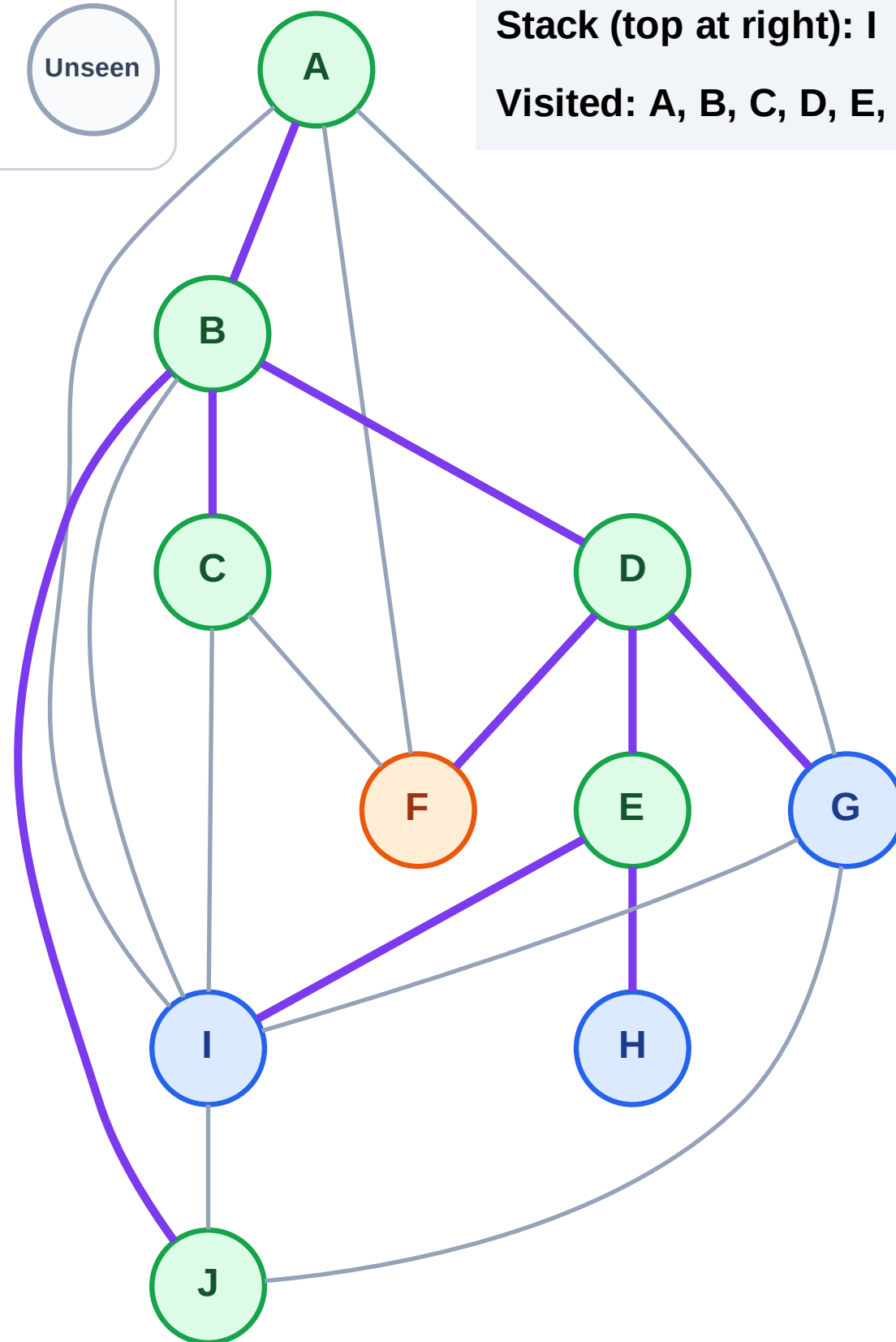
Step 14: Process node F

Legend



Stack (top at right): I | H | G

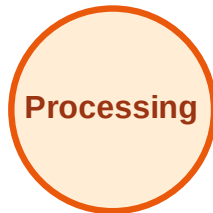
Visited: A, B, C, D, E, J



DFS Traversal

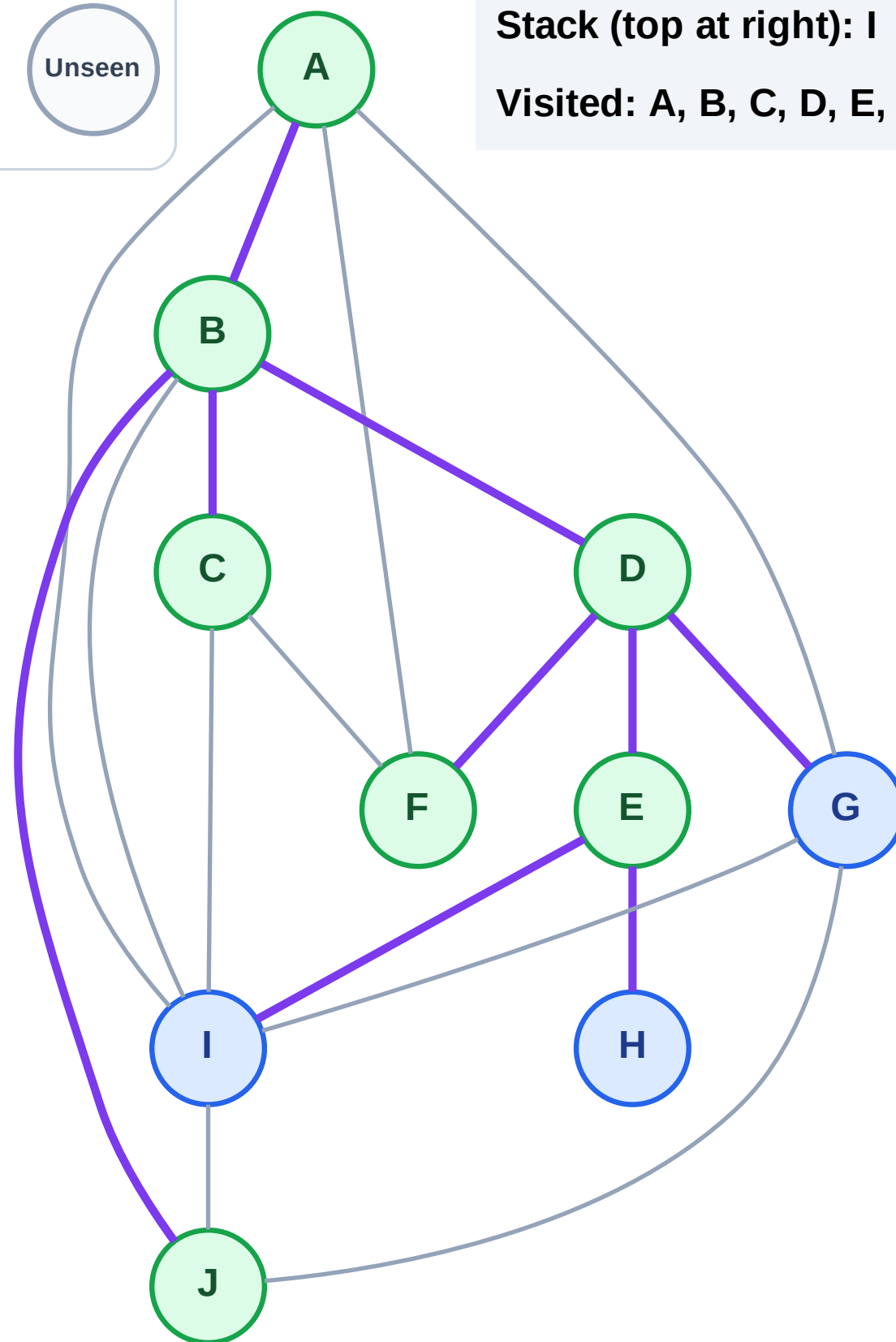
Step 15: No new neighbors from F

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

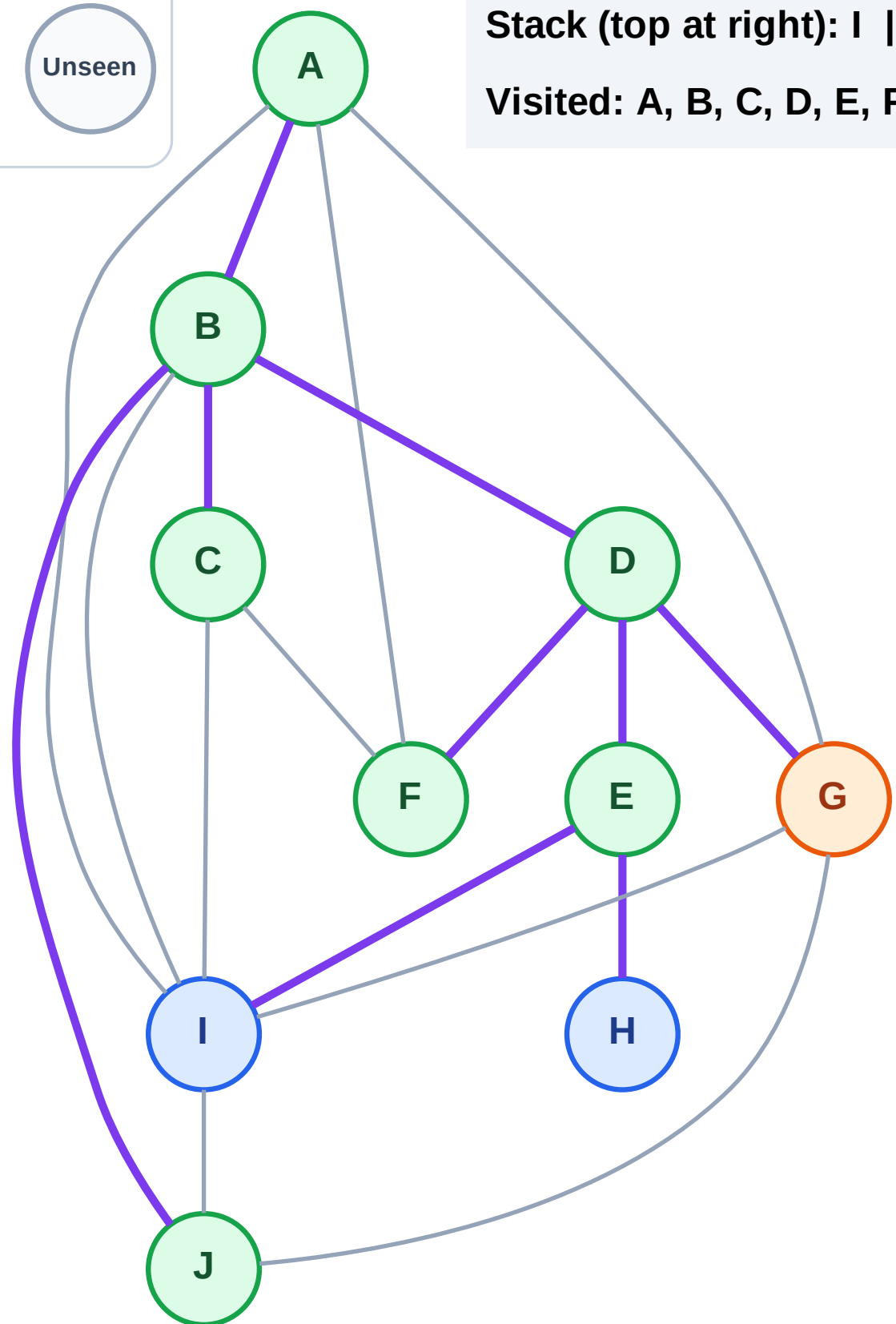
Processing

Discovered

Unseen

Stack (top at right): I | H

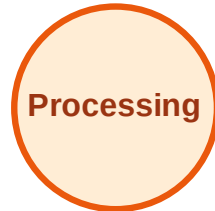
Visited: A, B, C, D, E, F, J



DFS Traversal

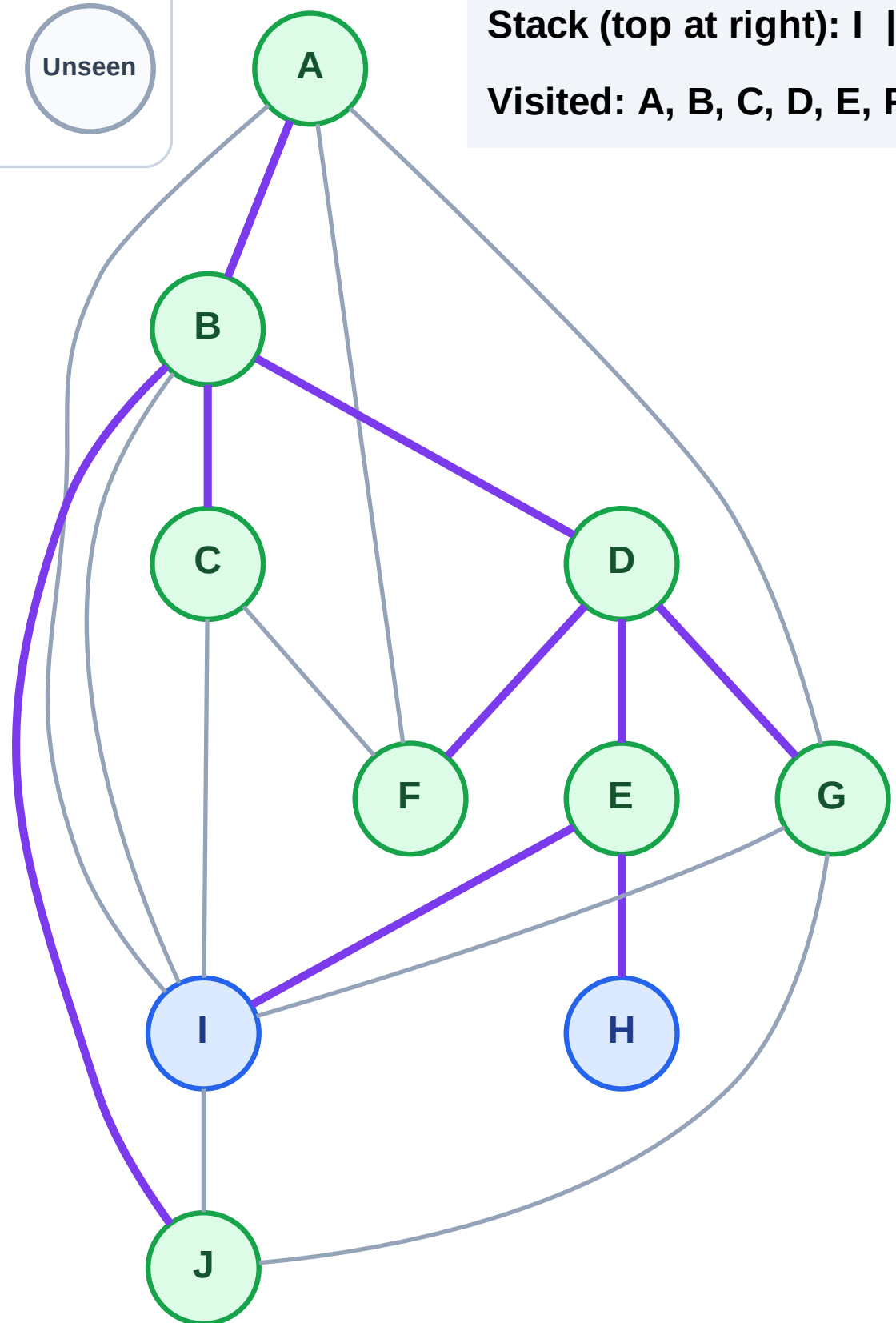
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

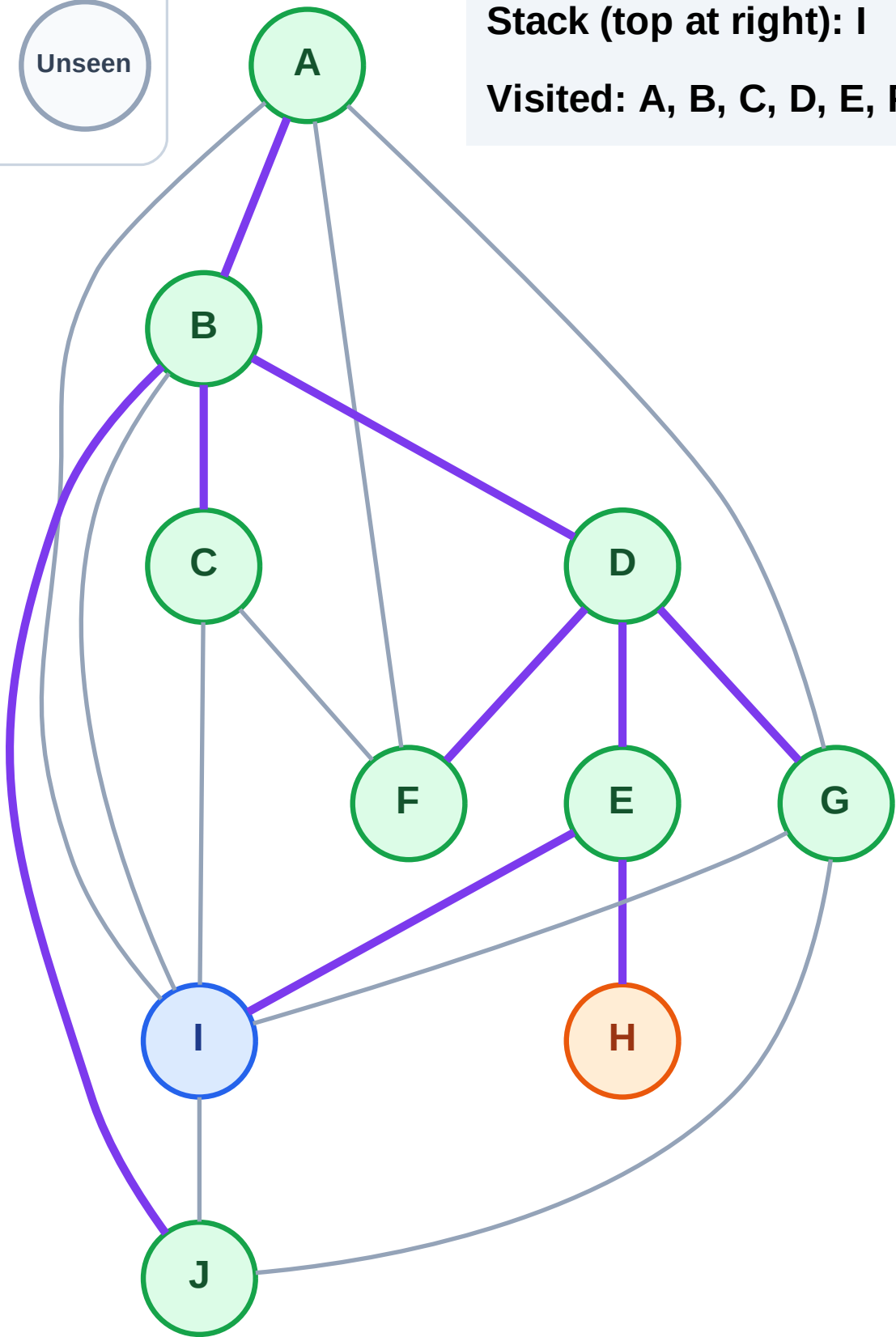
Complete

Processing

Discovered

Unseen

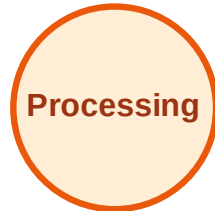
Stack (top at right): I
Visited: A, B, C, D, E, F, G, J



DFS Traversal

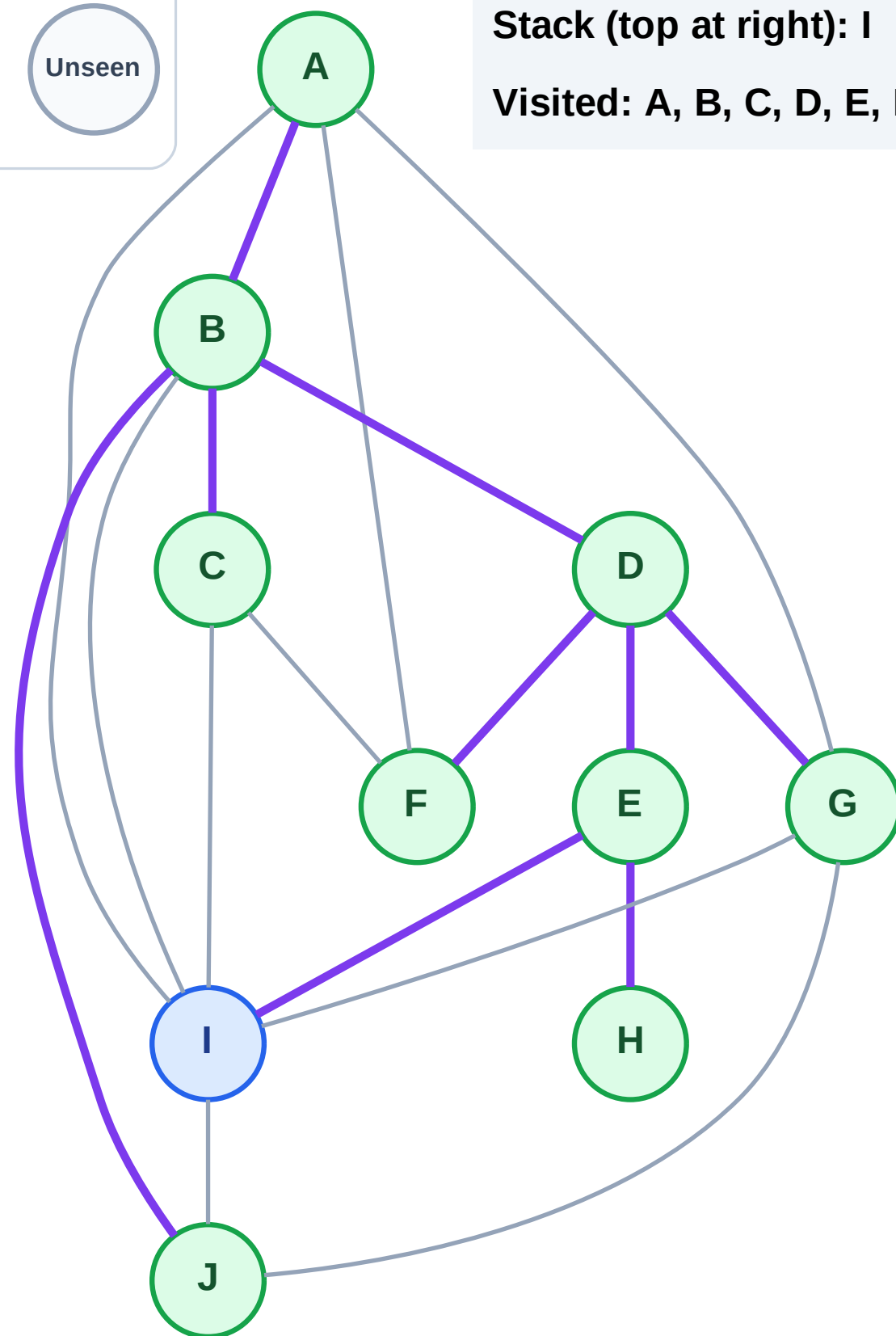
Step 19: No new neighbors from H

Legend



Stack (top at right): I

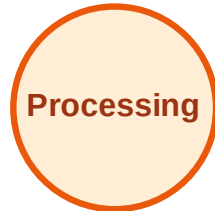
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

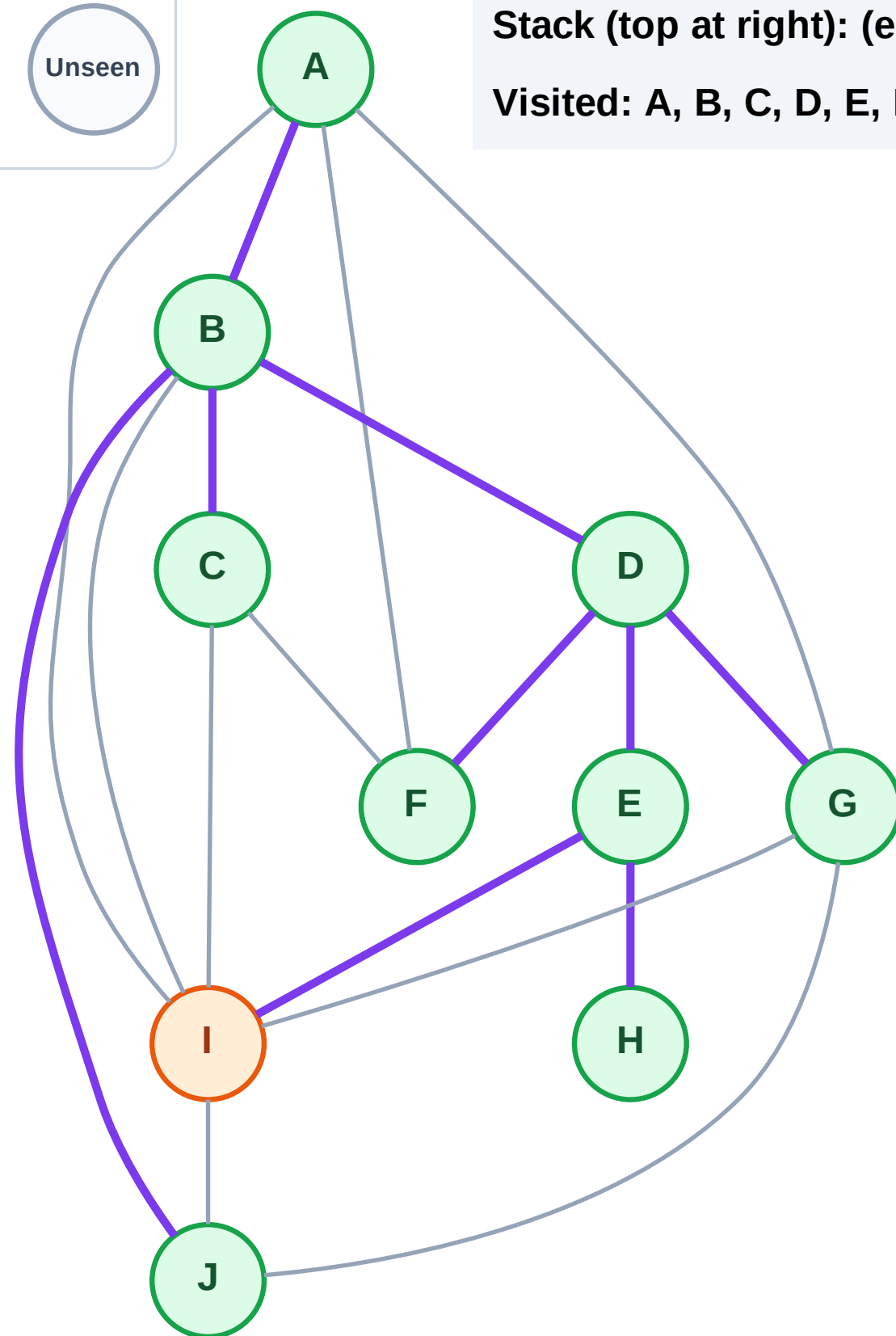
Step 20: Process node I

Legend



Stack (top at right): (empty)

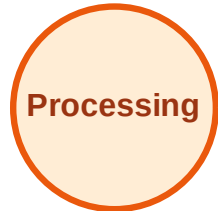
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

